

A Principled and Interpretable Orbit-Based Learning Framework

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Editor:

Abstract

We introduce a structural theory of learnability from an operator-theoretic perspective in which scaffold geometry, constructive realizability, and prediction are treated as linked structural bottlenecks rather than as disconnected stages of a pipeline. From this perspective, attention is not a primitive mechanism but a derived instance of a broader operator-theoretic principle. On the realizability side, we prove exact interpolation and realizability theorems, develop a budgeted scaffold-selection principle with certified projection-error control, and introduce CCDS, a constructive deviation-controlled synthesis mechanism yielding explicit and interpretable realizers on data-induced orbit scaffolds. On the predictive side, we show that the same structural logic extends to learning through Weighted Backward Shift Neural Networks (WBSNNs), whose predictions are obtained by learning coefficient laws over orbit dictionaries, thereby linking constructive realization and generalization within a single theory.

More broadly, the framework suggests a change of viewpoint for machine learning: datasets may carry intrinsic orbit geometry and geometric signatures, opening the door to dataset structure detection and topology-aware theories of learnability in which data geometry becomes as central as parametrization. The manuscript develops this theory in finite dimensions, shows that its main constructive mechanisms already extend to infinite-dimensional Banach settings, and supports the framework with empirical validation for both realizability and prediction across diverse datasets.

1 Introduction

Deep learning excels across vision, language, and time series, but much of its practical success still relies on stacked nonlinearities, heavy overparameterization, and architectural heuristics such as attention, which often obscure mechanism and demand large labeled datasets and careful tuning. In this work we ask a different question: can learning be organized around simple orbit dynamics, explicit geometry, and constructive realization rather than around increasingly opaque parametrization? We introduce an orbit-based framework in which scaffold geometry, realizability, and prediction are treated as three linked structural bottlenecks rather than as disconnected stages of a pipeline. The predictive instantiation of this framework, which we call a Weighted Backward Shift Neural Network (WBSNN), has a general-purpose design: the same orbit-based mechanism can be deployed across varied and demanding settings without requiring the pipeline to be tailored to each dataset. Moreover, although the backbone is linear, the model’s predictions become nonlinear through their dependence on orbit evolution and data-dependent coefficient selection, yielding both interpretability and expressive capacity.

The philosophy is the following. Instead of taking the model as the primary object and asking it to discover structure implicitly, we first extract from the data a structured orbit scaffold encoding part of the dataset’s intrinsic geometry. On that scaffold we study whether a target operator, or in the predictive setting a target dataset map, can be constructively realized with explicit and analyzable error. Prediction is then built on top of this constructive stage: rather than learning outputs freely, the predictor learns a coefficient law that tracks and extends the constructive rule

found on the training set. This leads to a framework organized around three explicit bottlenecks: a *scaffold bottleneck*, measuring how well the selected orbit scaffold captures the relevant geometry of the data; a *realizability bottleneck*, measuring how well the projected target can be constructively synthesized on that scaffold; and a *predictive coefficient bottleneck*, measuring how well the learned predictor reproduces and generalizes the constructive coefficient rule.

The theoretical contribution of the paper has two tightly connected parts. On the realizability side, we prove exact interpolation theorems showing that orbit-generated realizations exist under explicit structural conditions; we derive a three-term decomposition separating tail, projection, and parametrization errors; we develop an algorithm that searches for an optimal orbit-constrained scaffold; and we introduce CCDS, a constructive deviation-controlled synthesis mechanism that yields explicit certificates and recovers classical greedy pursuit as a boundary case. On the predictive side, we show that the same structural logic carries over to learning: the WBSNN predictor inherits the scaffold and realizability bottlenecks and adds only one new one, namely the learned approximation of the effective CCDS coefficient map. This yields in-sample structural prediction bounds and an out-of-sample transfer principle under geometric proximity assumptions between training and test points.

Why should the machine-learning community care about such a framework? First, it offers a principled alternative to black-box feature mixing: as a particular consequence of our realizability framework, attention is not a primitive mechanism, but rather a special case of a broader problem of realizing input-dependent linear operators on a data-induced orbit scaffold. Second, the framework is explicitly interpretable end-to-end: the scaffold is geometric, the realization step is constructive, and the predictor is constrained to generalize through orbit coefficients rather than through unrestricted parametrization. Third, the theory points toward a possible change of paradigm in which datasets are not merely collections of examples to be fitted, but objects with an intrinsic orbit geometry whose compatibility with operators can be probed through quantities such as the projection term. This suggests dataset signatures, geometric structure detectors, and, more broadly, a topology- or sheaf-oriented theory of learnability in which the geometry induced by the data becomes as central as the model itself. Finally, because both the exact interpolation theorem and the CCDS mechanism extend naturally to infinite-dimensional Banach settings, the framework opens a mathematically grounded route toward operator learning in PDEs, physics, control, inverse problems, climate and weather fields, and other settings where functions or operators, rather than finite vectors alone, are the native objects.

Our construction is rooted in the theory of linear dynamics, where the weighted backward shift operator plays a foundational role. In infinite-dimensional separable sequence Banach spaces, this operator acts on sequences $x = (x_0, x_1, x_2, \dots)$ by shifting coordinates backward and scaling by a bounded weight sequence $w = (w_i)_{i \geq 0}$:

$$B_w x = (w_1 x_1, w_2 x_2, w_3 x_3, \dots). \quad (1)$$

This simple mechanism gives rise to rich dynamical behavior, including dense orbits and hypercyclicity, and is therefore a natural source of structured representations. To adapt this dynamics to finite-dimensional learning problems, we introduce a wrapped finite-dimensional surrogate that cyclically reintroduces lost coordinates while preserving the shift-based orbit structure. This weighted backward shift mechanism serves as the backbone of the entire framework: it generates the orbit dictionary used to build scaffolds, supports the exact interpolation theory underlying realizability, and provides the structured representation class on which WBSNN performs prediction.

In this sense, the paper shifts the center of gravity of learning theory from parametrization alone to the interaction between dataset geometry, orbit structure, and constructive realization.

The framework’s backbone. At the core of our framework lies a structured, shift-based architecture $W^{(L)}$ (defined below), which cyclically shifts and scales input features across layers. Unlike traditional feedforward networks, its modulo- d design preserves all input dimensions throughout

depth, enabling persistent feature recombination. This recurrence serves as a finite-dimensional surrogate of orbit dynamics from infinite-dimensional operator theory, where iterated linear actions generate structured, dense trajectories. With only d parameters, $W^{(L)}$ builds global representations from local shifts, providing an interpretable and expressive backbone for both interpolation and generalization.

Definition 1 $W^{(m)}$ denotes a wrapped finite-dimensional iterate, used as a technical surrogate for backward-shift dynamics in infinite dimension: rather than shifting mass out of the ambient space, indices are taken modulo d , so that orbit iterates remain in $\mathbb{R}^d$ while still exhibiting shift-like propagation. Concretely, for $x \in \mathbb{R}^d$ and weights $w = (w_1, \dots, w_d) \in \mathbb{R}^d$, we set

$$(W^{(m)}x)_{i-1} := \left(\prod_{k=0}^{m-1} w_{(i+k) \bmod d} \right) x_{(i+m-1) \bmod d}, \quad (2)$$

for $i = 1, \dots, d-1$ and $m \geq 1$, with the empty product interpreted as 1 for $m = 0$.

This wrapping is not a standard finite-dimensional dynamical system; rather, it is a technical device that adapts shift-based orbit constructions, naturally defined in infinite-dimensional spaces, to a fixed finite dimension.

Note. For intuition, the reader may view $W^{(L)}x$ as a shift-and-scale transformation of x , beginning at the L -th entry and preserving the cyclic order of the components modulo d .

Example 1 Let $d = 4$, $L = 2$, and weights (w_0, w_1, w_2, w_3) . Then for an input vector $x = (x_0, x_1, x_2, x_3) \in \mathbb{R}^4$, then

$$W^{(2)}x = \begin{bmatrix} 0 & 0 & w_1 w_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & w_2 w_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_3 w_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & w_0 w_1 \end{bmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} w_1 w_2 x_2 \\ w_2 w_3 x_3 \\ w_3 w_0 x_0 \\ w_0 w_1 x_1 \end{pmatrix}.$$

Remark 2 This extension modulo d structure ensures that no information is lost, even for large L , and allows the model to propagate and recombine features across all positions. $W^{(L)}$ thus provides a structured linear backbone for learning with global receptive fields and low parameter complexity.

Example 2 (Weight regimes ($d = 3, L = 2$)) Let $w = (c, c, c)$, then $W^{(2)}x = (c^2 x_2, c^2 x_0, c^2 x_1)$. Thus, $c = 0.5$ (vanishing) $\Rightarrow$ decay by 0.5^2 ; $c = 1$ (neutral) $\Rightarrow$ pure permutation (no scaling); $c = 1.5$ (exploding) $\Rightarrow$ amplification by 1.5^2 . In general, the product $\prod_{k=0}^{d-1} w_k$ governs energy growth/decay, providing a simple knob for stability and expressivity.

The following lemma reveals a key structural property of W that limits the distinctiveness of large powers.

Lemma 3 Given W with d predictors, for every $X \in \mathbb{R}^d$ and every $L \geq 1$, we have that $W^{(L)}X = \lambda W^{(L \bmod d)}X$, where $\lambda = \left(\prod_{k=0}^{d-1} w_k \right)^m$ with $L = md + (L \bmod d)$.

Proof For $1 \leq i \leq d$, (2) gives $(W^{(L)}X)_i = \left(\prod_{k=0}^{L-1} w_{(i+k) \bmod d} \right) X_{\text{ext}, L, (i+L-1) \bmod d}$, where $X_{\text{ext}, L, j} = X_{j \bmod d}$ for all $0 \leq j \leq d + L - 1$. Similarly,

$$(W^{(L \bmod d)}X)_i = \left(\prod_{k=0}^{(L \bmod d)-1} w_{(i+k) \bmod d} \right) X_{\text{ext}, L \bmod d, (i+(L \bmod d)-1) \bmod d}.$$

Since $(i + L - 1) \bmod d = (i + (L \bmod d) - 1) \bmod d$, the indices match, so $X_{\text{ext}, L, (i+L-1) \bmod d} = X_{\text{ext}, L \bmod d, (i+(L \bmod d)-1) \bmod d}$. For $L \geq 1$, there exists $m \geq 0$ such that $L = md + (L \bmod d)$, where $0 \leq L \bmod d < d$. The product in $W^{(L)}$ splits as:

$$\prod_{k=0}^{L-1} w_{(i+k) \bmod d} = \left(\prod_{n=0}^{m-1} \prod_{k=0}^{d-1} w_{(i+nd+k) \bmod d} \right) \prod_{k=0}^{(L \bmod d)-1} w_{(i+md+k) \bmod d}.$$

Indeed, the first md terms of $\prod_{k=0}^{L-1} w_{(i+k) \bmod d}$ are m full cycles (of length d) with indices $i + nd + k$ for n from 0 to $m - 1$ and k from 0 to $d - 1$. So, for each n , we get

$$w_{(i+nd) \bmod d} \cdot w_{(i+nd+1) \bmod d} \cdot \dots \cdot w_{(i+nd+(d-1)) \bmod d}$$

which covers one full cycle of indices. The remaining $L \bmod d$ terms (from $k = md$ to $k = L - 1$) are

$$w_{(i+md) \bmod d} \cdot w_{(i+md+1) \bmod d} \cdot \dots \cdot w_{(i+md+(L \bmod d)-1) \bmod d}$$

which is exactly $\prod_{k=0}^{(L \bmod d)-1} w_{(i+md+k) \bmod d}$.

For each n , the inner product $\prod_{k=0}^{d-1} w_{(i+nd+k) \bmod d} = \prod_{k=0}^{d-1} w_{(i+k) \bmod d} = \prod_{k=0}^{d-1} w_k$, since the indices $(i + nd + k) \bmod d$ cycle through $i, i + 1, \dots, i + d - 1$. Thus, the first term is $\left(\prod_{k=0}^{d-1} w_k \right)^m$. Hence, $(W^{(L)}X)_i = \left(\prod_{k=0}^{d-1} w_k \right)^m (W^{(L \bmod d)}X)_i$, and the scalar $\lambda = \left(\prod_{k=0}^{d-1} w_k \right)^m$ satisfies the lemma. $\blacksquare$

2 Orbit-Based Constructive Realization of Linear Operators

Transformer attention is commonly presented as a distinctive mechanism whose expressive power derives from the softmax formula. In this work we adopt a different viewpoint: for a fixed token sequence $X = (x_1, \dots, x_n) \in \mathbb{R}^{d \times n}$, single-head attention induces a *data-dependent linear operator* $A(X) \in \mathbb{R}^{d \times d}$, and the model output is obtained by applying this operator to token representations. From the operator perspective, the essential question is not how $A(X)$ is computed, but whether a structured model can *constructively realize* the same operator action on the data, with explicit and analyzable error.

More concretely, given a linear operator $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$, a finite input-target dataset $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$, a fixed dictionary $\mathcal{D} \subset \mathbb{R}^C$, and a tolerance $\varepsilon > 0$, we ask whether there exists an element $\hat{y} : \mathbb{R}^d \rightarrow \mathbb{R}^C$ with $\{\hat{y}(x) : x \in \Xi\} \subset \text{span}(\mathcal{D})$ such that $\|A(\cdot) - \hat{y}(\cdot)\|_{\Xi} < \varepsilon$, where the empirical mean seminorm on Ξ is defined by $\|F\|_{\Xi} := \frac{1}{n} \sum_{i=1}^n \|F(x_i)\|_2$. This manuscript initiates a principled study of this realizability problem when the dictionary $\mathcal{D}$ is not arbitrary, but is a structured, data-induced orbit scaffold encoding the intrinsic geometry of Ξ (formalized below).

In the present work, we approach operator realizability via orbit dictionaries, and the first step towards this goal is to construct a data-induced orbit scaffold associated with a dataset Ξ .

Select a small anchor subset $\Xi_{\text{subset}} \subseteq \Xi$ and a partition $\{D_k\}_{k=1}^K$ of Ξ_{subset} . Phase 1 learns a weighted backward shift operator W . Phase 2 applies the exact interpolation Theorem 4 (stated below and proved in Appendix ??) to each block D_k , producing linear operators J_k that realize exact fitting on D_k .

The following is the foundational layer of the proposed framework. It shows that, under a technical independence condition, any finite dataset admits an exact realization within a class of transformed weighted shifts. We now present and prove an explicit construction of such a realization, which will serve as the foundation for our learning algorithm.

Theorem 4 (Exact Interpolation) Let W on $\mathbb{R}^d$ with weights $w_n \neq 0$, and let $\{(x_i, y_i)\}_{i=1}^{n_0} \subseteq \Xi_{\text{subset}} \subset \mathbb{R}^d \times \mathbb{R}^C$ with $n_0 \leq d$. Suppose for each $i = 1, \dots, n_0$, there exists $L_i \in \{0, \dots, d-1\}$ such that

$$W^{(L_i)}x_i \notin \text{span}\{W^{(L_1)}x_1, \dots, W^{(L_{i-1})}x_{i-1}\}. \quad (3)$$

Then, there exists a linear operator $J : \mathbb{R}^d \rightarrow \mathbb{R}^C$ such that $y_i = JW^{(L_i)}x_i$ for all $i = 1, \dots, n_0$.

Remark 5 On J as an Isomorphism: Define the invertibility metric $\delta := \max_{1 \leq i \leq N} \|Y_i - J_{i-1}W^{(L_i)}X_i\|$. By iterative construction, $J = I_d + \sum_{i=1}^N (Y_i - J_{i-1}W^{(L_i)}X_i) \otimes f_i^*$, so $\|J - I_d\| \leq N\delta \cdot \max_i \|f_i^*\|$. Thus, J is an isomorphism if $\|J - I_d\| < 1$ (see Lemma 2.1, p 192 (?)), achieved when $\delta < 1/(N \cdot \max_i \|f_i^*\|)$. The restriction $L_i \in \{0, \dots, d-1\}$ in Theorem 4 follows from Lemma 3, as $W^{(L)}X$ cycles with period d up to scaling. Beyond this, higher L only rescales earlier terms, adding no new linear independence information.

Consequently, for every training pair $(x_i, y_i) \in \Xi_{\text{subset}}$ with $x_i \in D_k$ there exists an integer $L_i \geq 1$ such that

$$y_i = J_k W^{(L_i)}x_i,$$

so that orbit iterates under W are mapped by J_k exactly onto the target outputs.

We then define the orbit-based scaffold $U(\Xi)$ relative to Ξ_{subset} as the following subset of $\mathbb{R}^C$

$$\left\{ J_k W^{(m)}x_i : (x_i, y_i) \in \Xi, 1 \leq k \leq K, 0 \leq m \leq d-1 \right\}.$$

The resulting scaffold $\mathcal{S}(\Xi) := \text{span } U(\Xi)$ is a finite-rank, data-induced subspace of the output Hilbert space that encodes the orbit geometry of the dataset.

Proof of Exact Interpolation Theorem 4 Let W on $\mathbb{R}^d$ with weights $w_n \neq 0$. Let also $(x_i, y_i)_{i=1}^{n_0} \subseteq \Xi_{\text{subset}} \subset \mathbb{R}^d \times \mathbb{R}^C$ satisfying (3).

Initialize: Set $J_0 = 0 : \mathbb{R}^d \rightarrow \mathbb{R}^C$.

Iterative construction: For $i = 1$. Pick any $L_1 \in \{0, \dots, d-1\}$ and find a linear functional $f_1^* \in (\mathbb{R}^d)^*$ with $f_1^*(W^{(L_1)}x_1) = 1$. Define $J_1 v := J_0 v + (y_1 - J_0 W^{(L_1)}x_1) \otimes f_1^*(v) = 0 + (y_1 - 0) \otimes f_1^*(v)$. Thus, $J_1 W^{(L_1)}x_1 = y_1 \cdot 1 = y_1$.

For $i \geq 2$: Assume

$$J_{i-1}W^{(L_j)}x_j = y_j \quad \text{for } j \leq i-1, \quad (4)$$

and pick $L_i \in \{0, \dots, d-1\}$ satisfying condition (3). By Hahn-Banach Theorem (see Corollary 3.15 on p.112 (7)), there exists a linear functional $f_i^* \in (\mathbb{R}^d)^*$ such that $f_i^*(W^{(L_j)}x_j) = 0$ for $j \leq i-1$, and $f_i^*(W^{(L_i)}x_i) = 1$. Define J_i as follows:

$$J_i v := J_{i-1}v + (y_i - J_{i-1}W^{(L_i)}x_i) \otimes f_i^*(v). \quad (5)$$

Thus, $J_i W^{(L_i)}x_i = J_{i-1}W^{(L_i)}x_i + (y_i - J_{i-1}W^{(L_i)}x_i) \cdot 1 = y_i$ and $J_i W^{(L_j)}x_j = J_{i-1}W^{(L_j)}x_j = y_j$, for $j \leq i-1$ since $f_i^*(W^{(L_j)}x_j) = 0$ and (4). By induction, for $1 \leq i \leq n_0$: $J_i W^{(L_j)}x_j = y_j$ for $j \leq i$. Finally, by setting $J := J_{n_0}$, we have that $Y_i = JW^{(L_i)}x_i$ for all $1 \leq i \leq n_0$. ■

Remark 6 The linear independence condition (3) is a sufficient condition for exact interpolation and serves as a clean theoretical certificate of realizability. In practice, however, we do not enforce it as a hard requirement in noisy or compressed regimes, and exact independence is not needed for stable performance. Phase 1 tests span membership via a least-squares residual, which we use as a practical proxy for independence, and Phase 2 employs regularized least-squares interpolation to handle rank deficiency and noise. This defines what we call the relaxed (non-exact) interpolation modality: exact fitting is intentionally sacrificed in favor of stability and robustness, and this is the regime used in all our main experiments.

Our main message is that attention is not fundamental but one instance of input-dependent linear mixing, and that our orbit-based framework provides a principled operator-theoretic mechanism to constructively realize such mixing. Attention is viewed as a black box producing an operator $A(X)$ and we study the realization of the map $\nu \mapsto A(X)\nu$ on the data subspace via orbit-based structure, without relying on softmax algebra, probabilistic arguments, or universality assumptions. We do not propose a new attention mechanism or an improved softmax approximation; rather, we view attention as a special case of the general problem of realizing finite-rank, input-dependent linear operators. This perspective exposes obstructions to certifiable realization and yields sufficient conditions for controlled operator realizability that are invisible in the standard attention formulation.

Definition 7 (Weighted Backward Shift Realizer) *Let $\hat{y} : \mathbb{R}^d \rightarrow \mathbb{R}^C$. The pair $(U(\Xi), \hat{y})$ is called a Weighted Backward Shift Realizer (WBSR) relative to Ξ_{subset} if $U(\Xi)$ is the orbit scaffold relative to Ξ_{subset} and $\hat{y}(x) \in U(\Xi)$ for all $x \in \Xi$.*

The following decomposition is an immediate consequence of the triangle inequality and is therefore purely structural. The technical core of this work is the constructive collapse of the parametrization term and the proof of existence of orbit scaffolds with small projection error; the algorithmic search for such optimal scaffolds is deferred to future work.

Theorem 8 (Three-term error decomposition for operator realization on an orbit scaffold)

Let $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$ be a finite dataset, $\Xi_{\text{subset}} \subseteq \Xi$, and let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be a linear operator. Let A_r denote its best rank- r approximation in spectral norm, with singular value tail $\sigma_{r+1}(A)$. Let $U(\Xi)$ be the orbit scaffold relative to Ξ_{subset} and $\Pi_{U(\Xi)} : \mathbb{R}^C \rightarrow \mathbb{R}^C$ the associated orthogonal projector.

Define the projected operator

$$\text{proxy}_{A, \Xi}(x) := \Pi_{U(\Xi)}(A_r x),$$

and let $(U(\Xi), \hat{y})$ be any WBSR relative to Ξ_{subset} . Then the following decomposition holds:

$$\begin{aligned} \|A - \hat{y}\|_{\Xi} &\leq \underbrace{\|A - A_r\|_{\Xi}}_{\varepsilon_{\text{tail}}} + \underbrace{\|A_r - \Pi_{U(\Xi)} A_r\|_{\Xi}}_{\varepsilon_{\text{proj}}} \\ &\quad + \underbrace{\|\Pi_{U(\Xi)} A_r - \hat{y}\|_{\Xi}}_{\varepsilon_{\text{param}}}. \end{aligned}$$

Moreover, by Eckart–Young,

$$\varepsilon_{\text{tail}} \leq \|A - A_r\|_2 \frac{1}{n} \sum_{i=1}^n \|x_i\|_2 \leq \sigma_{r+1}(A) \frac{1}{n} \sum_{i=1}^n \|x_i\|_2.$$

The meaning of this decomposition is not limited to an upper bound for a particular realization algorithm. The dataset Ξ induces, through Phases 1–2, a scaffold $U(\Xi)$ (a geometric “DNA” extracted from Ξ) that is a data-only object, independent of the operator A . The projection term $\varepsilon_{\text{proj}}(A, \Xi)$ depends only on A and on this dataset-induced scaffold, and measures how much of the action of A on Ξ lies in the orbit geometry intrinsic to Ξ . Equivalently, it quantifies how much of the structure inherent to the operator A is carried by the dataset Ξ . Small values indicate that the rank-truncated outputs $A_r x$ are largely contained in the subspaces encoded by $U(\Xi)$, while large values signal directions of A that are transversal to the dataset’s orbit structure. In particular, for fixed Ξ and varying A , the map $A \mapsto \varepsilon_{\text{proj}}(A, \Xi)$ can be interpreted as a probe of operator–dataset geometric compatibility, a viewpoint that suggests a dataset structure–detection principle which we do not pursue further here and leave to future work. To the best of our knowledge, this interpretation is new: we are not aware of prior work that formulates or exploits such a dataset–operator

structure detector, and we view it as a promising direction for future research. The parametrization term $\varepsilon_{\text{param}}(A, \Xi; \hat{y})$ instead measures whether, once this geometric compatibility holds, a WBSR constrained to $U(\Xi)$ can constructively synthesize the projected target. The decomposition therefore separates operator complexity, operator–dataset alignment, and algorithmic realizability, and can be read simultaneously as an approximation bound, a realizability statement, and a geometric diagnostic.

Theorem 8 identifies scaffold selection and realization as the two fundamental learning bottlenecks in orbit-based models. These two components will be the focus of the remainder of this manuscript.

We begin addressing both bottlenecks within this paper. On the geometric side, we formulate scaffold selection as an orbit-constrained optimization problem over a compact family of finite dimensional subspaces of $\mathcal{S}(\Xi)$, and prove existence of an optimal scaffold within the admissible orbit span. On the realization side, we adopt a deviation geometry perspective and introduce a certifiable deviation-controlled algorithm. When both projection error and parametrization error collapse, the linear operator can be constructively realized on the data subspace with arbitrarily small error. In this regime, the model does not merely imitate attention mechanisms, but constructively realizes the same input-dependent linear operators that attention produces. Low-rank attention thus appears as a particular instantiation of a more general operator realization principle.

2.1 Projection error and scaffold selection problem.

We first show that the existence of optimal scaffolds under which projection error collapses, is a mathematically well-posed problem. Denote by $\mathcal{U}$ the set of all scaffolds relative to Ξ .

Theorem 9 (Existence of a best orbit-constrained scaffold) *Let $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$, there exists $U_\star \in \mathcal{U}$ such that $\|A_r - \Pi_{U_\star} A_r\|_\Xi = \min_{U \in \mathcal{U}} \|A_r - \Pi_U A_r\|_\Xi$.*

Proof First, let us compute an upper bound for the dimension of any scaffold U relative to Ξ . Pick any subdataset $\Xi_{\text{sub}} \subset \Xi$ and assume it is partitioned as $\Xi_{\text{sub}} = D_0 \sqcup \dots \sqcup D_{K-1}$. Then, each sample $x_i \in \Xi_{\text{sub}}$ belongs to exactly one block D_k . Hence, in the dictionary construction, x_i contributes only the d atoms $J_k W^{(m)} x_i$, $0 \leq m \leq d-1$, associated with its own interpolant J_k . Therefore the total number of generating atoms is at most $d|\Xi_{\text{sub}}|$, and since the scaffold is a subspace of $\mathbb{R}^C$, we obtain $\dim U \leq \min(C, d|\Xi_{\text{sub}}|) \leq \min(C, dn)$.

If $\mathcal{U}_k$ denote the set of all k -dimensional scaffolds relative to Ξ . Hence,

$$\|A_r - \Pi_{U_\star} A_r\|_\Xi = \min_{1 \leq k \leq \min(C, dn)} \min_{U \in \mathcal{U}_k} \|A_r - \Pi_U A_r\|_\Xi.$$

In order to conclude our proof, it will be enough to prove the existence of a best orbit-constrained scaffold of dimension k , for all $1 \leq k \leq \min(C, dn)$. Since Ξ is finite, there are only finitely many admissible choices of subsets and partitions on which the exact interpolation theorem can be applied. Each such admissible choice produces one scaffold. Therefore, for each fixed k , the family of all k -dimensional scaffolds induced by Ξ is finite, hence compact. On the other hand, the map $U \mapsto \Pi_U$ is continuous (see, e.g., (7)). Define $f(U) = \|A_r - \Pi_U A_r\|_\Xi$. Since A_r is fixed and $U \mapsto \Pi_U$ is continuous in operator norm, f is continuous on $\mathcal{U}_k$. Because $\mathcal{U}_k$ is compact and f is continuous, the extreme value theorem implies that f attains its minimum on $\mathcal{U}_k$. Hence, there exists $U_\star^{(k)} \in \mathcal{U}_k$ such that $f(U_\star^{(k)}) = \min_{U \in \mathcal{U}_k} f(U) = \min_{U \in \mathcal{U}_k} \|A_r - \Pi_U A_r\|_\Xi$. $\blacksquare$

write better this pasticcio and come back to lemma before exact interp thm and finish in predictive parttttt Theorem 9 guarantees the existence of an optimal orbit-constrained scaffold, but it is non-constructive. Finding such a scaffold from data is not a classical approximation problem, because the admissible subspaces are not arbitrary: they must arise from orbit constructions induced by the learned operator and the geometry of Ξ . Scaffold selection is therefore a structured problem of its own.

To make this problem algorithmic, we isolate projection error and develop an operator-level CCDS theory for scaffold selection. The framework below yields a constructive nested-chain procedure, a deviation schedule that certifies the decay of projection error, and a feasibility criterion that identifies when a prescribed rate of collapse is impossible. We place this development before realizability for conceptual clarity: in the three-term decomposition, the scaffold must be selected before the realization step can be carried out. The following algorithm is the concrete outcome of this operator-level CCDS perspective.

To pass from existence to computation, we next develop a constructive theory for scaffold selection at the level of projection error itself. The goal is to produce a nested sequence of admissible orbit-constrained scaffolds together with quantitative control on how much the projection error decreases at each step. This leads to a certified selection procedure: it identifies when a prescribed rate of error collapse is achievable and when it is ruled out by the geometry of the admissible directions. Although this mechanism is conceptually parallel to the constructive solver introduced later for realizability, the two problems are different in nature: here the controlled object is projection error and the admissible moves are scaffold expansions, not coefficient updates. The algorithm below is the concrete outcome of this scaffold-level analysis.

Below is a rigorous “operator-level CCDS” formalization that mirrors your Phase-3 CCDS primitives (Lemma 4.3 / Theorem 4.4) but with the object being controlled equal to the projection error functional in your updated paper.

A. Setup: projection error as a matrix residual

A.1 Dataset seminorm and projection objective

Let $\Xi = \{x_i\}_{i=1}^N \subset \mathbb{R}^d$, and let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be linear. Let A_r be its best rank- r approximation (spectral norm) as in your paper.

Define the dataset seminorm

$$\|B\|_{\Xi} := \frac{1}{N} \sum_{i=1}^N \|Bx_i\|_2, \quad \text{or (more convenient for algebra) the squared variant} \quad \|B\|_{\Xi,2}^2 := \frac{1}{N} \sum_{i=1}^N \|Bx_i\|_2^2.$$

Your projection term is

$$\varepsilon_{\text{proj}}(A, \Xi) := \|A_r - \Pi_U A_r\|_{\Xi},$$

for a scaffold U (in $\mathbb{R}^C$) induced from Ξ .

For algorithmics, fix a particular X and orbit budget (W, M) and consider admissible subspaces

$$G_k(V_M(X)) := \{U \subset V_M(X) : \dim U = k\}, \quad f(U) := \|A_r - \Pi_U A_r\|_{\Xi},$$

exactly as in your scaffold selection subsection.

A.2 “Snapshot matrix” representation

Define the snapshot matrix of rank- r operator outputs

$$Y := [A_r x_1, \dots, A_r x_N] \in \mathbb{R}^{C \times N}.$$

Then for any orthogonal projector Π_U ,

$$\|A_r - \Pi_U A_r\|_{\Xi,2}^2 = \frac{1}{N} \sum_{i=1}^N \|(I - \Pi_U) A_r x_i\|_2^2 = \frac{1}{N} \|(I - \Pi_U) Y\|_F^2.$$

So, in the squared ℓ_2 dataset seminorm, projection error is exactly a Frobenius residual norm.

B. Operator-level CCDS primitive: one-step scaffold expansion and a “distance equation”

Phase-3 CCDS controls $\rho(r_t, Y_{t-1})$ along a nested chain $Y_0 \subset \dots \subset Y_T$ with 1-D increments; the key primitive is Lemma 4.3 (scalar distance equation) giving feasibility and explicit solutions.

For scaffold selection, the 1-D increment is not “choose a coefficient λ ” (as in Phase 3), but “choose a new direction u ” to enlarge the subspace. The right analogue of the “distance equation” becomes a statement about the achievable reduction in $\|(I - \Pi_U)Y\|_F$.

B.1 Nested admissible scaffold chain

Fix an ambient Hilbert space $H = \mathbb{R}^C$ with the standard inner product. Let $V \subset H$ be the admissible output scaffold space (e.g. $V = V_M(X)$ in your notation).

Let $U_0 \subset U_1 \subset \dots \subset U_T \subset V$ be a nested chain of subspaces with

$$\dim(U_t/U_{t-1}) = 1, \quad t = 1, \dots, T.$$

Write $U_t = U_{t-1} \oplus \text{span}\{u_t\}$ where $u_t \in V \cap U_{t-1}^\perp$ and $\|u_t\| = 1$.

Define the residual matrix at step t :

$$R_t := (I - \Pi_{U_t})Y, \quad E_t := \|R_t\|_F^2.$$

Lemma B.2 (One-step energy identity)

For $U_t = U_{t-1} \oplus \text{span}\{u\}$ with $u \perp U_{t-1}$, $\|u\| = 1$,

$$E_t = E_{t-1} - \|u^\top Y\|_2^2 = E_{t-1} - \|u^\top R_{t-1}\|_2^2.$$

Proof. Since $u \perp U_{t-1}$, $\Pi_{U_t} = \Pi_{U_{t-1}} + uu^\top$. Hence

$$R_t = (I - \Pi_{U_t})Y = (I - \Pi_{U_{t-1}} - uu^\top)Y = R_{t-1} - u(u^\top Y).$$

Then

$$\|R_t\|_F^2 = \|R_{t-1}\|_F^2 - 2\langle R_{t-1}, u(u^\top Y) \rangle_F + \|u(u^\top Y)\|_F^2.$$

But $\langle R_{t-1}, u(u^\top Y) \rangle_F = \text{tr}(R_{t-1}^\top u(u^\top Y)) = u^\top R_{t-1} Y^\top u = \|u^\top R_{t-1}\|_2^2$, and $\|u(u^\top Y)\|_F^2 = \|u\|^2 \|u^\top Y\|_2^2 = \|u^\top Y\|_2^2$. Combine terms to get the identity.

This is the scaffold analogue of the Phase-3 quadratic identity behind Lemma 4.3.

C. Operator-level “distance equation”: feasibility and explicit construction

In Phase-3 CCDS, for a fixed direction u , one solves for λ to enforce $\rho(r - \lambda u, Y) = d$ and feasibility is a scalar threshold $d \geq d_{\min}$.

Here, for a fixed current scaffold U , the controllable quantity is $\|u^\top R\|_2^2$ as u varies in the admissible complement.

Let $S := V \cap U^\perp$ and let $R := (I - \Pi_U)Y$. Define the quadratic form

$$q(u) := \|u^\top R\|_2^2 = u^\top (RR^\top)u, \quad u \in S, \quad \|u\| = 1.$$

Lemma C.1 (Feasible one-step residual energies)

Let $M := \Pi_S R R^\top \Pi_S$ (a symmetric PSD operator on H with range in S). Define

$$\lambda_{\min} := \min_{\substack{u \in S \\ \|u\|=1}} u^\top M u, \quad \lambda_{\max} := \max_{\substack{u \in S \\ \|u\|=1}} u^\top M u.$$

Then:

1. $q(u) = u^\top M u$ for all $u \in S$.
2. The set of achievable capture energies $\{q(u) : u \in S, \|u\| = 1\}$ equals the interval $[\lambda_{\min}, \lambda_{\max}]$.
3. Therefore the set of achievable next residual energies is exactly

$$E^+ \in [E - \lambda_{\max}, E - \lambda_{\min}], \quad \text{where } E = \|R\|_F^2.$$

Proof.

1. If $u \in S$, then $u = \Pi_S u$, and $u^\top R R^\top u = u^\top \Pi_S R R^\top \Pi_S u = u^\top M u$.
2. $u \mapsto u^\top M u$ is continuous on the compact unit sphere in S , so it attains min/max. Since the sphere in S is connected and the image of a connected set under a continuous function is connected, the image is an interval. Its endpoints are the min and max, hence $[\lambda_{\min}, \lambda_{\max}]$.
3. By Lemma B.2, $E^+ = E - q(u)$, so it ranges over $E - [\lambda_{\min}, \lambda_{\max}]$.

Interpretation. This is the exact scaffold analogue of CCDS feasibility:

- Phase-3: feasibility is $d \geq d_{\min}$ for the scalar distance equation.
- Scaffold-level: feasibility is “the desired energy drop lies in $[\lambda_{\min}, \lambda_{\max}]$.”

Lemma C.2 (Explicit 2D construction to hit a prescribed one-step drop)

Assume $\dim S \geq 2$ and let $v_1, v_2 \in S$ be orthonormal eigenvectors of M with eigenvalues $\lambda_1 \geq \lambda_2$ (top two). Then for any target $e \in [\lambda_2, \lambda_1]$, the choice

$$u(\theta) = \cos \theta v_1 + \sin \theta v_2, \quad \text{with } \cos^2 \theta = \frac{e - \lambda_2}{\lambda_1 - \lambda_2},$$

satisfies $q(u(\theta)) = e$, hence enforces $E^+ = E - e$.

Proof. Since $M v_j = \lambda_j v_j$,

$$u^\top M u = \cos^2 \theta \lambda_1 + \sin^2 \theta \lambda_2.$$

Choose $\cos^2 \theta$ as stated to make this equal to e .

This is the exact analogue of the “explicit solution” aspect of Lemma 4.3—except the scalar knob is now θ (mixing two eigen-directions) rather than λ .

D. Operator-CCDS Theorem (exact deviation enforcement for projection error)

Your Phase-3 Theorem 4.4 says: given a target deviation profile (d_t) satisfying per-step feasibility thresholds, there exists a deterministic sequence of coefficients (λ_t) achieving $\rho(r_t, Y_{t-1}) = d_t$ exactly. Here is the scaffold analogue.

Theorem D.1 (Constructive deviation-control principle for scaffold selection)

Let $V \subset \mathbb{R}^C$ be a closed subspace (admissible scaffold space). Let $Y \in \mathbb{R}^{C \times N}$ be fixed and set $R_0 := Y$, $U_0 := \{0\}$, $E_0 := \|R_0\|_F^2$.

Fix an integer $T \geq 1$ and a target energy profile

$$E_0 \geq E_1 \geq \dots \geq E_T \geq 0.$$

Define at each step t the admissible complement $S_{t-1} := V \cap U_{t-1}^\perp$ and the PSD operator

$$M_{t-1} := \Pi_{S_{t-1}} R_{t-1} R_{t-1}^\top \Pi_{S_{t-1}}.$$

Let $\lambda_{\min}^{(t-1)} \leq \lambda_{\max}^{(t-1)}$ be the min/max Rayleigh quotients of M_{t-1} on the unit sphere of S_{t-1} (as in Lemma C.1).

Assume the feasibility inequalities

$$E_{t-1} - E_t \in [\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}], \quad t = 1, \dots, T.$$

Then there exists a deterministic sequence of unit vectors $u_t \in S_{t-1}$ and a nested scaffold chain

$$U_t := U_{t-1} \oplus \text{span}\{u_t\} \subset V,$$

such that with $R_t = (I - \Pi_{U_t})Y$ one has the exact enforced constraints

$$\|R_t\|_F^2 = E_t, \quad t = 1, \dots, T.$$

Moreover (constructive choice). If $\dim S_{t-1} \geq 2$, one can pick u_t explicitly via Lemma C.2 using the top two eigenpairs of M_{t-1} to hit $E_{t-1} - E_t$ exactly.

Proof. Induct on t . Assume U_{t-1} and R_{t-1} are constructed with $\|R_{t-1}\|_F^2 = E_{t-1}$. By Lemma C.1, the achievable capture energies $q(u)$ for $u \in S_{t-1}$ lie in $[\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}]$. The feasibility assumption ensures there exists $u_t \in S_{t-1}$ such that

$$q(u_t) = E_{t-1} - E_t.$$

Define $U_t = U_{t-1} \oplus \text{span}\{u_t\}$. By Lemma B.2,

$$\|R_t\|_F^2 = \|R_{t-1}\|_F^2 - q(u_t) = E_{t-1} - (E_{t-1} - E_t) = E_t.$$

Proceed sequentially.

Corollary D.2 (Projection error deviation profile in dataset norm)

In the squared dataset seminorm,

$$\varepsilon_{\text{proj}}(U_t)^2 = \|A_r - \Pi_{U_t} A_r\|_{\Xi, 2}^2 = \frac{1}{N} \|R_t\|_F^2 = \frac{E_t}{N},$$

so Theorem D.1 enforces a prescribed projection error profile exactly.

E. The “spectral step” (greedy collapse) and its certified bound

If you do not want a prescribed profile, the canonical analogue of “choose $\lambda_t = \alpha_t$ to minimize distance” is:

Proposition E.1 (Greedy one-step maximal collapse)

At step t , among all unit $u \in S_{t-1}$, the choice maximizing the drop $q(u) = \|u^\top R_{t-1}\|_2^2$ is any top eigenvector of M_{t-1} ; the achieved drop equals $\lambda_{\max}^{(t-1)}$. Equivalently, it is the left singular vector associated with the largest singular value of $\Pi_{S_{t-1}} R_{t-1}$.

This is immediate from the Rayleigh–Ritz principle applied to M_{t-1} .

So your “CCDS-certified scaffold selection” can be presented as:

- certified feasibility for any desired drop sequence, via Theorem D.1;
- greedy maximal drop as a distinguished choice, via Proposition E.1.

2.2 Parametrization error and the role of WBSRs.

Once a scaffold subspace $\mathcal{S}(\Xi)$ has been fixed and the projection error $\varepsilon_{\text{proj}}$ controlled, the remaining approximation error lies in the realization of the compressed operator on that scaffold. The following example shows that exact span containment does not guarantee stable or efficient collapse of the parametrization error. In particular, a purely greedy innovation rule is insufficient: it has no mechanism to enforce a prescribed decrease of the residual norm, nor to prevent oscillatory selection among nearly collinear atoms. The realizability problem is therefore not solved by classical greedy pursuit alone, even when the projection error is zero. This pathology motivates deviation-controlled solvers such as our CCDS introduced here (all details below), which explicitly regulate the residual trajectory and stabilize the coefficient growth, ensuring constructive convergence even in the presence of severe geometric degeneracies of the scaffold.

Example 3 (Canonical ill-conditioned realizability example) Consider $\mathbb{R}^2$ with scaffold atoms $v_1 = (1, 0)^\top$, $v_2 = (\cos \theta, \sin \theta)^\top$, with $0 < \theta \ll 1$, and target $y = (0, 1)^\top$. Since $\text{span}\{v_1, v_2\} = \mathbb{R}^2$, the projection error vanishes and exact realization is possible: $\alpha_1 v_1 + \alpha_2 v_2 = y$. The unique solution is $\alpha_2 = 1/\sin \theta$ and $\alpha_1 = -\cos \theta / \sin \theta$, so $|\alpha_1| + |\alpha_2| \sim \theta^{-1} \rightarrow \infty$ as $\theta \rightarrow 0$. Thus exact realizability requires arbitrarily large coefficients with near-perfect cancellation.

Moreover, since the angle between v_1 and v_2 is θ , successive orthogonal projections alternate between the two atoms and the residual admits the exact closed form $\|r_t\| = \cos^t \theta$. In particular, for small θ one has $\cos \theta = 1 - \theta^2/2 + O(\theta^4)$, so $\|r_t\| = \cos^t \theta \approx \exp(-t\theta^2/2)$; hence a constant-factor reduction of the residual requires on the order of $1/\theta^2$ alternating steps as $\theta \rightarrow 0$. The corresponding coefficients satisfy $a_{2k+1} = \cos^{2k} \theta \sin \theta$ and $a_{2k+2} = -\cos^{2k+1} \theta \sin \theta$: they alternate in sign and are successively scaled by factors of $\cos \theta$, building up large cancelling contributions whose sum converges to the ill-conditioned exact coefficients $\alpha_1 = -\cos \theta / \sin \theta$ and $\alpha_2 = 1/\sin \theta$. This explicit geometric-series accumulation makes precise that exact realizability is achieved only through the delicate cancellation of two large, nearly equal contributions, rendering the parametrization extremely sensitive and explaining the observed coefficient blow-up in the near-collinear regime.

Geometrically, the atoms become almost collinear. Indeed, one has $\langle r_{2k}, v_1 \rangle = 0$ and $\langle r_{2k}, v_2 \rangle = a_{2k+1} = \cos^{2k} \theta \sin \theta$, hence $\langle r_{2k}, v_1 \rangle - \langle r_{2k}, v_2 \rangle = -a_{2k+1}$. Likewise, $\langle r_{2k+1}, v_2 \rangle = 0$ and $\langle r_{2k+1}, v_1 \rangle = a_{2k+2} = -\cos^{2k+1} \theta \sin \theta$, hence $\langle r_{2k+1}, v_1 \rangle - \langle r_{2k+1}, v_2 \rangle = a_{2k+2}$. Therefore the atoms are selected in strict alternation, and the correlation gap has magnitude $|a_{t+1}| = \cos^t \theta \sin \theta = O(\sin \theta)$, so the selection rule becomes arbitrarily sensitive to perturbations as $\theta \rightarrow 0$.

Example 3 shows that even with zero projection error, classical greedy pursuit is insufficient to guarantee stable collapse of parametrization error. Controlled innovation is required to prevent oscillation and coefficient blow-up, which motivates our deviation-controlled solvers defined below.

CCDS: Constructive Deviation-Controlled Synthesis for Phase 3. Let $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ be a real Hilbert space with norm $\|x\| = \sqrt{\langle x, x \rangle}$. For any closed subspace $Y \subset \mathcal{H}$ let Π_Y denote the orthogonal projector and

$$\rho(x, Y) := \inf_{y \in Y} \|x - y\| = \|x - \Pi_Y x\|$$

the distance from x to Y .

Nested scaffolds. Fix a nested chain of closed subspaces $Y_0 \subset Y_1 \subset \dots \subset Y_T \subset \mathcal{H}$ with $\dim(Y_t/Y_{t-1}) = 1$ for $t = 1, \dots, T$.¹ For each t pick a unit *innovation direction* $u_t \in Y_t \cap Y_{t-1}^\perp$ (hence $Y_t = Y_{t-1} \oplus \text{span}\{u_t\}$).

Deviation profile. Given a target $y \in \mathcal{H}$, CCDS will construct an approximation $\hat{y} \in Y_T$ via a sequence $\hat{y}_0, \hat{y}_1, \dots, \hat{y}_T$ with $\hat{y}_{t-1} \in Y_{t-1}$ and residuals $r_{t-1} := y - \hat{y}_{t-1}$. The key quantity controlled at step t is the *deviation of the updated residual from the previous scaffold*:

$$d_t := \rho(r_t, Y_{t-1}) \quad (t = 1, \dots, T), \quad (6)$$

1. In applications, one either enforces $\dim(Y_t/Y_{t-1}) = 1$ by construction (e.g. one atom per step), or one works with a block version. The 1-dimensional increment case is the canonical CCDS primitive.

where $r_t := y - \hat{y}_t$ after the step- t update. The idea is: at each refinement step we add a new direction, so we can prescribe (within feasibility) how far the new residual remains from the preceding scaffold.

Scalar distance equation. Fix t and abbreviate $Y := Y_{t-1}$, $u := u_t \in Y^\perp$ and $r := r_{t-1}$. Consider the scalar function $\phi(\lambda) := \rho(r - \lambda u, Y) = \|(I - \Pi_Y)(r - \lambda u)\| = \|a - \lambda u\|$, $a := (I - \Pi_Y)r \in Y^\perp$. This is continuous, strictly convex in λ (in squared form), and its level sets can be solved exactly.

Lemma 10 (Distance equation: explicit solutions and feasibility) *Let $Y \subset \mathcal{H}$ be closed, let $u \in Y^\perp$ with $\|u\| = 1$, and let $r \in \mathcal{H}$. Set $a := (I - \Pi_Y)r \in Y^\perp$ and $\alpha := \langle a, u \rangle$. Then for any target $d \geq 0$, the equation*

$$\rho(r - \lambda u, Y) = d \quad (7)$$

has a real solution $\lambda \in \mathbb{R}$ if and only if

$$d \geq d_{\min} := \inf_{\lambda \in \mathbb{R}} \rho(r - \lambda u, Y) = \sqrt{\|a\|^2 - \alpha^2}. \quad (8)$$

When feasible, (7) has exactly two solutions

$$\lambda_{\pm} = \alpha \pm \sqrt{d^2 - d_{\min}^2}. \quad (9)$$

Moreover, the unique minimizer is $\lambda^ = \alpha$, attaining $\rho(r - \lambda^* u, Y) = d_{\min}$.*

Proof Since $u \in Y^\perp$ and $\|u\| = 1$,

$$\begin{aligned} \rho(r - \lambda u, Y)^2 &= \|a - \lambda u\|^2 \\ &= \|a\|^2 - 2\lambda \langle a, u \rangle + \lambda^2 \\ &= (\lambda - \alpha)^2 + (\|a\|^2 - \alpha^2). \end{aligned}$$

Thus $\rho(r - \lambda u, Y) = d$ is equivalent to $(\lambda - \alpha)^2 = d^2 - (\|a\|^2 - \alpha^2)$. Real solutions exist iff $d^2 \geq \|a\|^2 - \alpha^2$, i.e. $d \geq d_{\min}$, yielding (9). The minimizer is the vertex $\lambda^* = \alpha$ with minimal value $d_{\min}$. $\blacksquare$

Constructive principle (finite-horizon CCDS). Lemma 10 yields a fully constructive, step-wise principle:

Theorem 11 (Constructive deviation-control principle) *Fix a nested chain $Y_0 \subset \dots \subset Y_T$ with $\dim(Y_t/Y_{t-1}) = 1$ and innovation directions $u_t \in Y_t \cap Y_{t-1}^\perp$ with $\|u_t\| = 1$. Let $y \in \mathcal{H}$ and initialize $\hat{y}_0 = 0 \in Y_0$, $r_0 = y$. Fix any target deviation profile $(d_t)_{t=1}^T$ satisfying, for each t ,*

$$\begin{aligned} d_t \geq d_{t,\min} &:= \sqrt{\|a_t\|^2 - \alpha_t^2}, \quad a_t := (I - \Pi_{Y_{t-1}})r_{t-1}, \\ \alpha_t &:= \langle a_t, u_t \rangle. \end{aligned} \quad (10)$$

Then, there exists a (deterministic) sequence of coefficients $(\lambda_t)_{t=1}^T$ such that, defining

$$\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t \in Y_t, \quad r_t := y - \hat{y}_t,$$

one has for every $t = 1, \dots, T$ the exact deviation constraints $\rho(r_t, Y_{t-1}) = d_t$. The coefficients admit the explicit formula $\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}$, with the sign chosen by any deterministic rule.

Proof Fix t . By construction $u_t \in Y_{t-1}^\perp$, and r_{t-1} is known from previous steps. The feasibility condition (10) is exactly (8) in Lemma 10 with $Y = Y_{t-1}$, $u = u_t$, $r = r_{t-1}$ and target $d = d_t$. Hence there exists λ_t solving $\rho(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$. Setting $\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t$ gives $r_t = y - \hat{y}_t = r_{t-1} - \lambda_t u_t$ and thus $\rho(r_t, Y_{t-1}) = d_t$. Proceed sequentially for $t = 1, \dots, T$. $\blacksquare$

CCDS algorithm. We now define CCDS as an algorithmic Phase-3 solver that instantiates the constructive principle above on a chain induced from a dictionary.

Dictionary and ordering. Let $\mathcal{D} \subset \mathcal{H}$ be a unit-norm dictionary. Fix a deterministic ordering rule that, given a state (Y_{t-1}, r_{t-1}) , selects a new atom $q_t \in \mathcal{D}$ such that $q_t \notin Y_{t-1}$. Define

$$u_t := \frac{q_t - \Pi_{Y_{t-1}} q_t}{\|q_t - \Pi_{Y_{t-1}} q_t\|} \in Y_{t-1}^\perp, \quad Y_t := Y_{t-1} \oplus \text{span}\{u_t\}. \quad (11)$$

Let $\eta_t := \|q_t - \Pi_{Y_{t-1}} q_t\| > 0$, and let $e_t \in \mathbb{R}^t$ denote the t -th canonical basis vector. Define recursively coefficient vectors $c^{(t)} \in \mathbb{R}^t$ by $c^{(1)} := (1)$, and, for $t \geq 2$,

$$c^{(t)} := \frac{1}{\eta_t} \left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \tilde{c}^{(s,t)} \right), \quad (12)$$

where $\tilde{c}^{(s,t)} \in \mathbb{R}^t$ is the zero-padded extension of $c^{(s)}$, namely $\tilde{c}^{(s,t)} := (c_1^{(s)}, \dots, c_s^{(s)}, 0, \dots, 0)$. Then set

$$\beta^{(t)} := (\beta^{(t-1)}, 0) + \lambda_t c^{(t)} \in \mathbb{R}^t. \quad (13)$$

Proposition 12 (Equivalent selected-atom representation of CCDS output) *For each $t = 1, \dots, T$ one has $u_t = \sum_{j=1}^t c_j^{(t)} q_j$, and consequently $\hat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{j=1}^T \beta_j^{(T)} q_j$.*

Proof We argue by induction on t . For $t = 1$, since $Y_0 = \{0\}$, one has $u_1 = q_1$, so $c^{(1)} = (1)$ and the claim holds. Assume now that for each $s < t$, $u_s = \sum_{j=1}^s c_j^{(s)} q_j$. Since $\{u_s\}_{s=1}^{t-1}$ is an orthonormal basis of Y_{t-1} , we have $\Pi_{Y_{t-1}} q_t = \sum_{s=1}^{t-1} \langle q_t, u_s \rangle u_s$. Substituting the inductive representation of each u_s ,

$$q_t - \Pi_{Y_{t-1}} q_t = q_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \sum_{j=1}^s c_j^{(s)} q_j.$$

After zero-padding the vectors $c^{(s)}$ to $\mathbb{R}^t$, this becomes

$$q_t - \Pi_{Y_{t-1}} q_t = \sum_{j=1}^t \left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \tilde{c}^{(s,t)} \right)_j q_j.$$

Dividing by $\eta_t = \|q_t - \Pi_{Y_{t-1}} q_t\|$ yields $u_t = \sum_{j=1}^t c_j^{(t)} q_j$. This proves the first claim. Finally,

$$\hat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{t=1}^T \lambda_t \sum_{j=1}^t c_j^{(t)} q_j = \sum_{j=1}^T \beta_j^{(T)} q_j,$$

by the recursive definition of $\beta^{(t)}$. This proves the result. ■

Definition 13 (CCDS algorithm with explicit atom-dictionary output) *Fix a horizon T and a deviation profile rule that provides targets $d_t \geq 0$ at each step. Given an input target $y \in H$, define $\hat{y}_0 := 0$, $Y_0 := \{0\}$, $r_0 := y$, and initialize the dictionary-coefficient vector by $\beta^{(0)} := 0 \in \mathbb{R}^0$. For $t = 1, \dots, T$:*

1. *Select $q_t \in \mathcal{D}$ deterministically and form u_t, Y_t by (11).*
2. *Compute $a_t := (I - \Pi_{Y_{t-1}}) r_{t-1}$, and $\alpha_t := \langle a_t, u_t \rangle$. Define $d_{t,\min} := \sqrt{\|a_t\|^2 - \alpha_t^2}$.*
3. *Choose a target deviation $d_t \geq d_{t,\min}$ (feasibility).*

4. Set the CCDS coefficient by the explicit root $\lambda_t := \alpha_t - \sqrt{d_t^2 - d_{t,\min}^2}$, (or the $+$ root, or any deterministic root-selection rule).
5. Update $\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t$ and $r_t := y - \hat{y}_t$.
6. Dictionary-coordinate update. Let $c^{(t)}$ and $\beta^{(t)}$ defined as in (12) and (13) respectively.

The intrinsic CCDS output is $\hat{y}_T \in Y_T$. Equivalently, the same output admits the explicit selected-atom representation

$$\hat{y}_T = \sum_{j=1}^T \beta_j^{(T)} q_j \in \text{span}\{q_1, \dots, q_T\} \subset \text{span}(D).$$

Finite-dimensional infeasibility. The following example shows that in finite dimension, scaffold exhaustion can force u_t to be nearly orthogonal to the residual direction a , producing a strictly positive feasibility threshold $d_{\min}$. Certain deviation targets d_t are then geometrically impossible.

Example 4 (Finite-dimensional infeasibility and necessity of dimension lifting) Consider a single CCDS step with ambient Hilbert space $H = \mathbb{R}^2$, current scaffold subspace $Y_{t-1} = \{0\}$, one data point $x_1 = e_1 = (1, 0)$, target operator $A = I$, so the residual at step $t-1$ is $r_{t-1}(x_1) = Ax_1 = e_1$, and off-scaffold component $a := (I - \Pi_{Y_{t-1}})r_{t-1}(x_1) = e_1$, with $\|a\| = 1$.

Assume that the current scaffold dictionary is exhausted in the sense that the only available innovation direction is $u_t = e_2 = (0, 1)$, so that the update direction is forced to lie in the one-dimensional subspace $\mathcal{U}_t = \text{span}\{e_2\} \subset Y_{t-1}^\perp$. Hence, Lemma 10 (feasibility bound) gives

$$d_{\min} = \sqrt{\|a\|^2 - \langle a, u_t \rangle^2} = \sqrt{1 - 0} = 1,$$

so for all target deviations $d_t < 1$ the CCDS step is infeasible in the current finite-dimensional realization space: no choice of λ can achieve the target deviation because the available innovation direction is orthogonal to the off-scaffold component a .

Adding one new orthogonal dimension restores feasibility by allowing a new innovation direction with controlled alignment to a . This justifies the need for dimension lifting in order to guarantee universal deviation feasibility and, consequently, collapse of the parametrization error in CCDS.

Theorem 14 (One-step feasibility by dimension lifting) Let H be a real Hilbert space, $Y \subset H$ a closed subspace, and $r \in H$. Define $a := (I - \Pi_Y)r \in Y^\perp$. Fix a target deviation d satisfying $0 \leq d \leq \|a\|$. Then there exists a Hilbert space $\tilde{H} := H \oplus \mathbb{R}$, an isometric embedding $\iota : H \rightarrow \tilde{H}$, and a unit vector $u \in (\iota(Y))^\perp \subset \tilde{H}$ such that the CCDS distance equation $\rho(\iota(r) - \lambda u, \iota(Y)) = d$ admits a real solution $\lambda \in \mathbb{R}$. Moreover, one can choose u so that the minimal achievable deviation is exactly $d_{\min}(u) = d$.

Proof Define $\tilde{H} := H \oplus \mathbb{R}$ with inner product

$$\langle (x, s), (y, t) \rangle_{\tilde{H}} := \langle x, y \rangle_H + st,$$

and the isometric embedding $\iota(x) = (x, 0)$. Then $\iota(Y)$ is a closed subspace and

$$(\iota(Y))^\perp = Y^\perp \oplus \mathbb{R}.$$

Let

$$\tilde{r} := \iota(r), \quad \tilde{a} := (I - \Pi_{\iota(Y)})\tilde{r} = (a, 0),$$

so that $\|\tilde{a}\| = A$. If $A = 0$, then $d = 0$ and the claim is trivial. Assume $A > 0$ and define $e_1 := \frac{\tilde{a}}{A}$, $e_2 := (0, 1)$. Then $e_1, e_2 \in (\iota(Y))^\perp$ are orthonormal. Choose $\theta \in [0, \pi/2]$ such that $\sin \theta = \frac{d}{A}$. Define $u := \cos \theta e_1 + \sin \theta e_2$. Then $u \in (\iota(Y))^\perp$ and $\|u\| = 1$. Moreover, $\alpha := \langle \tilde{a}, u \rangle_{\tilde{H}} = A \cos \theta$.

By the CCDS feasibility formula (Lemma 3.3),

$$d_{\min}(u) = \sqrt{\|\tilde{a}\|^2 - \alpha^2} = \sqrt{A^2 - A^2 \cos^2 \theta} = A \sin \theta = d.$$

Hence $d \geq d_{\min}(u)$, and the equation $\rho(\tilde{r} - \lambda u, \iota(Y)) = d$ admits a real solution λ . This completes the proof. $\blacksquare$

Remark 15 *To enforce the same target deviation level d simultaneously for M sample-wise residuals $\{r^{(j)}\}_{j=1}^M$ at a given CCDS step, it suffices to iterate the one-step dimension-lifting construction of Theorem 14 for each residual. This yields an isometric embedding into an ambient Hilbert space of dimension at most $\dim(H) + M$ in which the deviation constraint $\rho(r^{(j)} - \lambda u, Y) = d$ is feasible for all $j = 1, \dots, M$.*

Resolution of Example 4 by lifting one dimension. Embed H isometrically into $\tilde{H} = \mathbb{R}^3$ by $x \mapsto (x, 0)$, and extend the orthogonal complement with the new basis vector e_3 . Choose the lifted innovation direction $\tilde{u}_t = \cos \theta e_1 + \sin \theta e_3$, $\sin \theta = d_t$. Then $\tilde{u}_t \in (\iota(Y_{t-1}))^\perp$, $\|\tilde{u}_t\| = 1$, and $\langle \tilde{a}, \tilde{u}_t \rangle = \cos \theta$, $d_{\min}(\tilde{u}_t) = \sqrt{1 - \cos^2 \theta} = \sin \theta = d_t$. Hence the CCDS equation

$$\rho(\iota(r_{t-1}(x_1)) - \lambda \tilde{u}_t, \iota(Y_{t-1})) = d_t$$

admits a real solution, and the deviation d_t becomes feasible after adding a single orthogonal dimension.

Greedy residual-alignment as the boundary case of CCDS. At Phase 3, CCDS constructs updates of the form $r_t = r_{t-1} - \lambda_t u_t$, where $u_t \in Y_{t-1}^\perp$ is an innovation direction and λ_t is selected by solving a one-dimensional distance equation $\rho(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$, for a feasible deviation target $d_t \geq d_{t,\min}$.

Consider the trivial scaffold $Y = \{0\}$. Then $\Pi_Y = 0$ and $\rho(r, Y) = \|r\|$. Innovation directions coincide with dictionary atoms: $u = q \in \mathcal{D}$. The CCDS distance equation reduces to $\|r_{t-1} - \lambda_t q_t\| = d_t$, with feasibility interval

$$d_t \in \left[\min_{\lambda} \|r_{t-1} - \lambda q_t\|, \|r_{t-1}\| \right].$$

Lemma 16 is a classical fact from Hilbert space geometry (orthogonal projection onto a one-dimensional subspace). We include a short proof for completeness and to make the paper self-contained.

Lemma 16 (Minimal-deviation update) *For fixed $r \in \mathcal{H}$ and unit $q \in \mathcal{H}$, the function $\lambda \mapsto \|r - \lambda q\|^2$ attains its unique minimum at $\lambda^* = \langle r, q \rangle$, with minimum value $d_{t,\min} = \|r\|^2 - |\langle r, q \rangle|^2$.*

Proof Let $r \in H$ and let $q \in H$ with $\|q\| = 1$. For $\lambda \in \mathbb{R}$ we expand

$$\|r - \lambda q\|^2 = \langle r - \lambda q, r - \lambda q \rangle = \|r\|^2 - 2\lambda \langle r, q \rangle + \lambda^2.$$

This quadratic polynomial in λ attains its minimum at $\lambda^* = \langle r, q \rangle$. Evaluating at this value gives $\min_{\lambda \in \mathbb{R}} \|r - \lambda q\|^2 = \|r\|^2 - \langle r, q \rangle^2$. Hence the optimal coefficient is $\lambda^* = \langle r, q \rangle$ and the minimal deviation is $d_{\min} = \sqrt{\|r\|^2 - \langle r, q \rangle^2}$. $\blacksquare$

Thus, choosing the deviation target $d_t = d_{t,\min}$ and selecting the corresponding minimizing root yields the update $r_t = r_{t-1} - \langle r_{t-1}, q_t \rangle q_t$, and the CCDS framework collapses into the Matching Pursuit algorithm. We refer to this Phase-3 realization regime as the *Matching Pursuit regime (MP)*. Thus, MP is defined as the specialization of CCDS to the trivial scaffold $Y = \{0\}$ (no nested subspace scaffolds) with minimal deviation targets $d_t = d_{t,\min}$, in which the general deviation equation reduces to greedy contraction and geometric residual decay.

Remark 17 *MP therefore appears as the boundary case of a deviation–equation solver on nested-subspaces orbit scaffolds.*

So we have.

Proposition 18 *MP is exactly CCDS restricted to the trivial scaffold $Y = \{0\}$ with minimal-deviation target selection.*

Additionally, under a uniform alignment condition, MP yields geometric residual decay. This is the classical Matching Pursuit contraction result; for completeness, we provide a self-contained detailed exposition of this fact in the Appendix ??.

Interpretability of WBSRs. WBSRs are fully interpretable end-to-end: Phase 1 learns an explicit shift operator W , Phase 2 yields verifiable linear maps J_k by interpolation, and Phase 3 constructs sparse realizations via deviation-controlled CCDS with geometric certificates. All components are transparent, with no black-box nonlinear mixing or probabilistic normalization, enabling direct inspection of the algebraic and geometric mechanisms of operator realization, in contrast to standard attention models which are typically analyzed only through input–output behavior.

2.3 Realizability in an Operator-Theoretic Machine Learning Context.

Residual thresholds are classical in inverse problems, most notably through Morozov’s discrepancy principle and its variants, where a target residual level is used as a stopping or regularization-parameter selection criterion rather than as an equality constraint at each iteration (14; 2). By contrast, the CCDS framework introduced here treats the residual-norm schedule $\{d_t\}$ as a first-class object: each greedy update enforces a prescribed deviation $\rho(r_{t+1}, Y_t) = d_{t+1}$ at every step by solving an explicit one-dimensional distance equation. This internalizes the deviation schedule into the update rule itself, rather than using it merely as a termination or tuning condition.

On the other hand, greedy methods such as Matching Pursuit (MP) (12) and Orthogonal Matching Pursuit (OMP) (15; 17) are classical tools for sparse approximation: they select atoms sequentially and determine coefficients via residual minimization or orthogonal projection. In parallel, the theory of greedy approximation in Banach and Hilbert spaces—developed extensively by Temlyakov and collaborators—formalizes broad families of greedy schemes (weak greedy algorithms, relaxed and free-relaxation variants) where updates follow descent inequalities or relaxation principles, with convergence analyzed a posteriori through rates or bounds (8; 16). In all these approaches, the per-step update is defined implicitly via an optimization or projection objective. **mention berstein lethargy theory**

CCDS recasts greedy selection as the constructive realization of a prescribed deviation profile rather than the optimization of a surrogate objective, a perspective that, to the best of our knowledge, does not appear in prior greedy pursuit or approximation algorithms. Moreover, the CCDS framework recovers the classical greedy (MP) update as the specialization to the trivial nested scaffold $Y_0 = \{0\}$ with one-dimensional extensions and extremal root selection of the distance equation. In this regime, innovation directions coincide with the selected atoms, and choosing the minimizing solution of the scalar deviation equation yields the standard orthogonal contraction along the best-aligned atom. Prior greedy pursuit methods treat maximal monotone contraction as the default primitive; here it emerges as a special case of controlled deviation. This MP/CCDS dichotomy provides a unified constructive solver that interpolates between deviation profiles under alignment and controlled deviation, complementing existing greedy methods that focus on approximation quality but do not explicitly encode deviation schedules at the algorithmic step level.

Consequently, our CCDS framework constitutes a novel conceptual contribution in three primary ways: (i) it is the first to treat the residual path as a prescribed constraint to be solved for, rather than a result to be monitored; (ii) it replaces implicit optimization with explicit synthesis; and (iii) it provides a unified theory that recovers standard greedy algorithms as a specialized subset of

a broader deviation-control class. Our work bridges these by grounding realization in orbit dynamics, providing the first constructive, verifiable framework for operator realizability in data-induced scaffolds—extending greedy theory to prescribed deviations and explicit low-rank realizations.

2.4 Experiments on realizability

merge at the end with all experiments together

Phase 3 Realizer Protocol (Experiments). In Phase 3, a fast greedy solver is applied on the trivial scaffold ($Y_0 = \{0\}$, without nested subspaces): at each step the most correlated atom is selected and a MP update is performed when $d_{\min} \leq \epsilon$; otherwise a CCDS distance-equation update is used with $d_t \in [d_{\min}, \|r_t\|]$. The procedure falls back to the full nested-subspace CCDS solver when the mean parametrization residual ϵ_{param} exceeds a preset tolerance (0.05 in our experiments). This fallback is an implementation-level safeguard and does not affect the realizability theory, which is independent of the particular deviation schedule.

Remark (Role of $d_t > d_{\min}$ in CCDS). Choosing $d_t > d_{\min}$ is not merely a numerical convenience. It (i) allows the prescription of arbitrary feasible convergence profiles beyond the extremal greedy collapse $d_t = d_{\min}$ (which forces $\lambda_t = \alpha_t$), (ii) moves the update away from the degenerate boundary of the distance equation, yielding numerically stable roots and robust orthogonalization, and (iii) improves the conditioning of successive innovation directions in nested realizations by preventing repeated near-collinearity. The extreme choice $d_t = d_{\min}$ gives the fastest contraction, but the freedom $d_t > d_{\min}$ is what endows CCDS with universality, stability, and well-conditioned scaffold growth. This freedom is exploited in the following experiments, particularly when alignment is weak or absent.

Experiment 1: Realizability under severe compression We evaluate the MP/CCDS realizability solver on Swiss roll, Gas Sensor Drift (6 classes), and Isolet (26 classes) using only raw or PCA-compressed inputs (no auxiliary features) and random, non-optimized Phase 1–2 scaffolds, so that all structure is induced solely by scaffold geometry.

On Swiss roll, two variants probe collapse of the parametrization error (ϵ_{param}): a noisy 3-class setting (75,000 samples, $\sigma = 0.5$, three quantile labels) tests broad class separation in $\mathbb{R}^3$; and a low-sample label-noise setting (1,275 samples, $\sigma = 0.1$, ten bins, 10% uniform flips, stratified) controls variance under extreme scarcity. Four representative configurations show: near-perfect collapse ($\epsilon_{\text{param}} \approx 6.8 \times 10^{-12}$) with extreme sparsity (mean support ≈ 1) and strong interpretability at sufficient budget; vacuous realizations at extreme low budgets ($\epsilon_{\text{proj}} = 1$, $\epsilon_{\text{param}} = 0$, no atoms); and robust low residuals ($\approx 8 \times 10^{-8}$) with graceful degradation under label noise and scarcity.

On Gas Sensor Drift (PCA $d = 5$, 89% variance, 5–7% scaffold), we obtain $\epsilon_{\text{param}} \approx 5 \times 10^{-8}$ with $\epsilon_{\text{proj}} \approx 3 \times 10^{-7}$ and high sparsity (98–99%), confirming exact constructive realization. At $d = 15$ with tiny budgets, $\epsilon_{\text{proj}} = 1$ while $\epsilon_{\text{param}} = 0$, isolating failure as purely geometric.

On Isolet (PCA $d = 5$, 42% variance), ϵ_{param} reaches $\approx 10^{-8}$ once the random scaffold is sufficiently expressive (e.g. at 3%, with $\epsilon_{\text{proj}} \approx 6 \times 10^{-7}$), while ϵ_{proj} falls sharply from 3.8×10^{-2} (1%) to 6×10^{-7} (3%), showing accuracy is controlled by scaffold geometry rather than solver limits. At $d = 35$ (75% variance) with tiny scaffolds, as well as at $d = 5$ with 10% budget, we observe the same vacuous projection regime ($\epsilon_{\text{proj}} = 1$). Both cases arise from unlucky random scaffold draws that induce a subspace nearly orthogonal to the one-hot labels, yielding $\epsilon_{\text{proj}} = 1$ and trivial realization.

Overall, MP/CCDS achieves near-perfect constructive realizability, fast certified convergence, and strong interpretability whenever the scaffold is geometrically adequate, while all observed failures are attributable to projection limitations rather than to the optimization or realization procedure itself—motivating adaptive and optimized scaffold construction. More details in Table ??.

Experiment 2: Pathological weak-alignment stress test and deviation-control ablation. We evaluate Phase-3 realizability on a synthetic dataset constructed to destroy global alignment: each sample is generated in an independent random orthogonal frame and re-rotated,

with strong label noise, yielding residuals in generic position relative to any fixed orbit dictionary. This places the solver in a small-cosine, thin-feasible-cone regime where monotone MP contraction is not guaranteed.

A Phase-1/2 scaffold is frozen under a strict budget ($d = 10$, $M = 700$, random 10% subset, operator family ill-conditioned). Interpolation is deliberately relaxed to accommodate noise, so Phase-2 leaves a dominant projection error ($\varepsilon_{\text{proj}} \approx 0.64$), and Phase-3 must perform genuine innovation.

An ablation over the deviation-control parameter $\gamma_{\text{dev}} \in \{0, 0.5, 0.9\}$ shows that the constructive realizability rate remains stable at $\approx 36\text{--}37\%$, indicating that the expressivity ceiling is determined by scaffold geometry rather than by the deviation schedule. In contrast, solver stability depends strongly on γ_{dev} : small values enforce $d_t \approx d_{\text{min}}$ and yield near-optimal parametrization error ($\varepsilon_{\text{param}} \sim 10^{-7}$) with the fast MP/CCDS solver, while large γ_{dev} produces thin feasible caps, stalls greedy alignment, and requires fallback to full CCDS. The latter recovers the same realizability rate, confirming that deviation control governs numerical stability and contraction, not the intrinsic realizable set. Residuals of order 10^{-2} under full CCDS reflect the imposed deviation schedule and finite step budget, not the minimal achievable error within the fixed scaffold. More details in Tables ?? and ??.

Despite the adversarial geometry, the solutions remain highly interpretable: representations are extremely sparse (median support size $|S_i| = 2$, global sparsity $> 94\%$) and supported on a small, stable subset of orbit atoms with consistent class-conditional usage. Most non-realizable samples terminate by reaching the step limit rather than by infeasibility, showing that the dominant limitation is scaffold incompleteness and projection geometry, not solver instability. More details in Table ??.

Experiment 3: Extreme-Ray Dominance vs. Interior Feasibility in Near-Parallel Subspaces under Weak Scaffolds. We evaluate Phase-3 realizability of MP/CCDS on a challenging synthetic dataset formed by a union of six nearly parallel 3-dimensional subspaces in $\mathbb{R}^{35}$ ($\sigma_{\text{parallel}} = 0.02$, $\sigma_{\text{noise}} = 0.05$), reduced to $d = 15$ by PCA (98.2% variance retained). Labels are generated by linear operators C followed by a fixed readout B , yielding classification tasks with identical input geometry but diverse operator spectra (identity, orthogonal, ill-conditioned, low-rank, spiked, sparse, Toeplitz, and adversarially scaffold-misaligned). A deliberately weak, unoptimized random scaffold (3% budget) is used to decouple Phase-3 realizability from scaffold selection.

Despite poor projection coverage, fast MP/CCDS consistently achieves extremely small parametrization error ($\varepsilon_{\text{param}} \sim 10^{-8}\text{--}10^{-6}$) via very sparse representations. However, strict MP contraction is almost never certified: residuals typically lie on flat or symmetric faces of the feasible cone, where several atoms have nearly equal correlations and no single extreme ray has a margin. Full CCDS, enforcing deviation-controlled multi-atom feasibility rather than one-step contraction, achieves moderate to high realizability (4–88%) across spectra, constructing stable interior two-atom certificates even under large projection error (e.g., low-rank case). More details on full diagnostics and interpretability metrics in Tables ?? – ??.

In particular, Tables ?? and ?? report the fast MP/CCDS and full CCDS traces for Sample 0 (Identity and Orthogonal), illustrating the sharp contrast between greedy dynamics and deviation-controlled certificates. In fast mode, the Identity case converges in 14 steps with occasional CCDS fallbacks, while the Orthogonal case requires 50 steps and exhibits strong alternation between a few atoms (e.g., repeated switching between 327 and 194), a clear instance of boundary crawling along a flat cone face. This directly visualizes the margin-free pathology: no single atom dominates, so pure MP progresses slowly and fragily. In contrast, full CCDS always produces a clean two-step interior-feasible certificate. Together, these traces provide concrete evidence that greedy contraction is insufficient in near-parallel geometries, whereas deviation control yields short, stable, and geometrically robust realizations.

Overall, Experiment 3 exposes a core limitation of classical greedy contraction: MP requires strict single-ray dominance and fails almost universally due to margin collapse under symmetry, degeneracy, or near-parallelism. Realizability is instead governed by cone interior width and operator-induced anisotropy. This benchmark cleanly separates approximation quality from contraction ge-

ometry and shows that deviation-controlled solvers such as CCDS are essential — not optional — for achieving reliable, stable realizability under weak, unoptimized scaffolds.

For a full report of interpretability diagnostics (per-sample supports, atom usage histograms, deviation paths, and stopping statistics), we refer the reader to the complementary GitHub repository accompanying this manuscript.

Vacuous regimes observed with random scaffolds disappear under optimized scaffold construction, since the resulting subspace can be deliberately aligned with the label directions, preventing projection collapse and enabling non-trivial sparse realizations. This directly motivates our future-work direction on optimized scaffold selection.

3 Predictive Instantiation of our Framework

The philosophy before formalization. Before entering the technical development of the predictive part of our work, we explain the guiding idea behind it. Let $\Xi_{train} = \{(x_i, y_i) : 1 \leq i \leq n\}$ be the training set. In our framework, the training set plays two distinct roles: first, it provides the target outputs that the predictor must fit; second, through a constructive coefficient assignment attached to the training inputs, it provides a preferred orbit-based representation of those outputs. This second signal is the central point of the framework.

A natural first attempt would be to use the entire training set as the interpolation set. Concretely, one could seek a partition $\Xi_{train} = D_0 \sqcup \dots \sqcup D_{K-1}$ such that the exact interpolation theorem applies on each block D_k . In that regime, one would learn a weighted shift W and interpolants J_k such that for every $(x_i, y_i) \in \Xi_{train}$ there exist k and m with $y_i = J_k W^{(m)} x_i$. This would provide an exact constructive teacher on the training set, and the predictive rule would then take the form

$$\tilde{f}(x) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}(x) J_k W^{(m)} x, \quad x \in \mathbb{R}^d.$$

However, this full-training regime is not the one we ultimately want. Since the entire training set is used to build the orbit dictionary, the coefficient map $x \mapsto \alpha(x) \in \mathbb{R}^{Kd}$ becomes highest-dimensional, the model moves closer to memorization, and the generalization step becomes harder rather than easier. Moreover, in noisy settings, exact interpolation of the full training set is undesirable because it encourages the model to fit noise. Thus, this regime is well suited to exact training realization, but not necessarily to robust prediction.

This observation leads naturally to the alternative adopted in this work: rather than building the scaffold from the whole training set, we build it from a carefully selected subset. On that subset, we again apply exact interpolation as many times as needed, learning a shift W and interpolants J_k . The corresponding terms $J_k W^{(m)} x$, with $(x, y) \in D_k$, form the orbit atoms of a dictionary $\mathcal{D}$, and their span $U = \text{span } \mathcal{D}$ defines the associated scaffold. We then select the subset whose scaffold is optimal in the sense that it captures the relevant output geometry of the full training set.

Once such a scaffold has been selected, the next step is not to interpolate the entire target directly, but to project the target onto the scaffold and construct on it a deterministic CCDS realizer. This yields an effective coefficient map on the training set. In other words, the CCDS realizer provides a constructive surrogate of the projected target on the scaffold. The final step is then to learn, on the full training set, a coefficient map that extends this constructive orbit-based rule from the training points to all of $\mathbb{R}^d$. Thus, the predictor is not trained to learn the target map freely. Rather, it is trained to reproduce and generalize the constructive coefficient rule induced by the CCDS realizer on the selected scaffold.

This is what makes the predictive step fundamentally different from ordinary supervised fitting. The training labels y_i tell the model what output to produce, but the CCDS realizer tells it how to produce that output through a preferred constructive orbit-based decomposition. In this sense,

the framework has a teacher-student flavor, except that the teacher is not a black box: it is the deterministic CCDS solver itself.

This compressed-scaffold regime has major advantages over the full-training interpolation regime. It yields smaller capacity, stronger inductive bias, better data efficiency, and a better chance of robust generalization, precisely because the predictor is heavily constrained. The price we pay is that the CCDS realizer approximates the projected target $\Pi_U f$, rather than the exact target on the whole training set. This is why scaffold selection becomes fundamental: the scaffold must be chosen optimally so that $\Pi_U f$ captures as much as possible of the relevant geometry of the full dataset.

Our predictive philosophy is therefore organized around three explicit structural bottlenecks: the scaffold bottleneck, the realizability bottleneck, and the predictive coefficient bottleneck. The first determines how well the selected scaffold captures the target geometry; the second measures how well the projected target can be constructively realized on that scaffold; and the third governs how well the learned predictor reproduces and extends the constructive coefficient rule beyond the training set. This is precisely what makes the framework unusual: the pieces genuinely talk to each other. The realizability theory developed in the first part of the paper is not isolated from the predictive part introduced here, but is carried over into it as its structural backbone, through the scaffold bottleneck, the CCDS realizability bottleneck, and the learned coefficient law that extends this constructive structure beyond the training set.

Formalization. One of the most striking features of our theory is that the structural decomposition developed for realizability carries over directly to prediction on the dataset. Since in the predictive setting the target map is simply the dataset correspondence $f(x_i) = y_i$, there is not separate spectral tail term on Ξ . Theorem 19 below is an in-sample structural theorem that identifies the right bottlenecks for learning on the training data: (i) the quality of the selected scaffold U , through the projection error $\|f - \Pi_U f\|_\Xi$; (ii) the quality of the CCDS realization on that scaffold through the parametrization error; and (iii) the quality of the shallow learned policy through its approximation on the effective CCDS coefficient map on the dataset.

Let $\Xi_{train} = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$, $\Xi_{test} = \{(\bar{x}_j, \bar{y}_j)\}_{j=1}^m \subset \mathbb{R}^d \times \mathbb{R}^C$. Define the dataset seminorm $\|g\|_\Xi := (\sum_{x:(x,y) \in \Xi} \|g(x)\|_{\ell^2})/|\Xi|$ for maps g defined on $\{x : (x, y) \in \Xi\}$.

Theorem 19 (Geometric in-sample prediction bound) *Fix a scaffold $U = \text{span } \mathcal{D} \subset \mathbb{R}^C$ with set of orbit atoms $\mathcal{D} = \{J_k W^{(m)} x_i : (x_i, y_i) \in \Xi_{train}, x_i \in D_k, 0 \leq k \leq K-1, 0 \leq m \leq d-1\}$. Define the target map on Ξ_{train} by $f(x_i) := y_i$. And let $f_{CCDS}(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} a_{k,m}^{\text{eff}}(x) u_{k,m}(x)$, with $x \in U$, be the CCDS realizer on the scaffold U as defined in [connect with previous section on realizability, insert definition like this instantiation there at the end of the section as an example to connect here more easily to the reader](#), where $\{u_{k,m}\}_{0 \leq k \leq K-1, 0 \leq m \leq d-1}$ is the family of innovation directions, with $u_{k,m} \neq 0$ for the indices selected by CCDS as defined in [\(??\)](#) and $u_{k,m} = 0$ for all remaining indices. Denote $a^{\text{eff}}(x) := (a_{k,m}^{\text{eff}}(x))_{k,m} \in \mathbb{R}^{Kd}$. Assume our predictor has the form*

$$\tilde{f}(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}(x) u_{k,m}(x)$$

and satisfies, for each $i = 1, \dots, n$, $\|\alpha(x_i) - a^{\text{eff}}(x_i)\|_{\ell^1} \leq \delta_i$. Then

$$\|f - \tilde{f}\|_{\Xi_{train}} \leq \underbrace{\|f - \Pi_U f\|_{\Xi_{train}}}_{\varepsilon_{proj}(f, U, \Xi_{train})} + \underbrace{\|\Pi_U f - f_{CCDS}\|_{\Xi_{train}}}_{\varepsilon_{param}(f_{CCDS}, U, \Xi_{train})} + M_u \bar{\delta},$$

where $\bar{\delta} := \frac{1}{n} \sum_{i=1}^n \delta_i$ and $M_u := \max_{0 \leq k \leq K-1, 0 \leq m \leq d-1, 1 \leq i \leq n} \|u_{k,m}(x_i)\|_{\ell^2} < \infty$ by boundedness of the operators $u_{k,m}$.

Proof Fix $i \in \{1, \dots, n\}$. Then

$$(\Pi_U f)(x_i) - \tilde{f}(x_i) = ((\Pi_U f)(x_i) - f_{CCDS}(x_i)) + (f_{CCDS}(x_i) - \tilde{f}(x_i)).$$

Hence, by the triangle inequality,

$$\|(\Pi_U f)(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq \|(\Pi_U f)(x_i) - f_{\text{CCDS}}(x_i)\|_{\ell^2} + \|f_{\text{CCDS}}(x_i) - \tilde{f}(x_i)\|_{\ell^2}.$$

We estimate the second term. By definition,

$$f_{\text{CCDS}}(x_i) - \tilde{f}(x_i) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} (a_{k,m}^{\text{eff}}(x_i) - \alpha_{k,m}(x_i)) u_{k,m} x_i.$$

Applying the triangle inequality again,

$$\|f_{\text{CCDS}}(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} |a_{k,m}^{\text{eff}}(x_i) - \alpha_{k,m}(x_i)| \|u_{k,m} x_i\|_{\ell^2}.$$

Then,

$$\|f_{\text{CCDS}}(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq M_u \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} |a_{k,m}^{\text{eff}}(x_i) - \alpha_{k,m}(x_i)|.$$

That is, $\|f_{\text{CCDS}}(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq M_u \|\alpha(x_i) - a^{\text{eff}}(x_i)\|_{\ell^1} \leq M_u \delta_i$. Therefore,

$$\|(\Pi_U f)(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq \|(\Pi_U f)(x_i) - f_{\text{CCDS}}(x_i)\|_{\ell^2} + M_u \delta_i.$$

Averaging over $i = 1, \dots, n$, we obtain

$$\|\Pi_U f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|\Pi_U f - f_{\text{CCDS}}\|_{\Xi_{\text{train}}} + M_u \bar{\delta}.$$

Finally, observe that $f - \tilde{f} = (f - \Pi_U f) + (\Pi_U f - \tilde{f})$. Applying the triangle inequality,

$$\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|f - \Pi_U f\|_{\Xi_{\text{train}}} + \|\Pi_U f - \tilde{f}\|_{\Xi_{\text{train}}}.$$

Substituting the bound just proved yields

$$\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|f - \Pi_U f\|_{\Xi_{\text{train}}} + \|\Pi_U f - f_{\text{CCDS}}\|_{\Xi_{\text{train}}} + M_u \bar{\delta}.$$

This completes the proof. ■

Theorem 19 shows that the in-sample predictive instantiation of our orbit-based framework, which we call Weighted Backward Shift Neural Networks (WBSNNs) and define formally below in Definition 24, is naturally governed by three bottlenecks that define the three phases of the model. The first is the construction of an optimal scaffold, which determines the projection quality and therefore defines Phase 1. The second bottleneck, and hence Phase 2, is the CCDS realizability of the projected targets on that scaffold. Phase 3 is the shallow learned policy that approximates the effective CCDS coefficient map on the dataset, thereby controlling the final parametrization gap. In this sense, prediction in WBSNN inherits the same structural logic as realizability, but with an additional learned coefficient-approximation layer in the final phase.

3.1 Predictive Framework (WBSNNs).

We now formalize WBSNN Phases 1–3.

Phase 1: Budgeted predictive scaffold selection in WBSNN. Let $\Xi_{train} = \{(x_i, y_i)\}_{i=1}^n$ be the training set. For $b \in \mathcal{B} \subset (0, 1]$, set $m_b := \lfloor bn \rfloor$ and $\mathfrak{X}_b := \{\Xi_{sub} \subset \Xi_{train} : |\Xi_{sub}| \leq m_b\}$.

Fix $\Xi_{sub} \in \mathfrak{X}_b$, write $M := |\Xi_{sub}|$, and let $c \in \mathcal{C} \subseteq \{1, \dots, M\}$. A *cap-constrained admissible partition* of Ξ_{sub} is a decomposition $\Xi_{sub} = D_0 \sqcup \dots \sqcup D_{K-1}$ such that $1 \leq |D_k| \leq c$ for all k , the recursive independence condition of Theorem ?? holds inside each D_k , and $K-1$ is minimal among such partitions; denote this minimum by $K(\Xi_{sub}, c) - 1$. The blocks need not be homogeneous.

Given (Ξ_{sub}, c) , apply exact interpolation Theorem or via regularized/pseudoinverse on each D_k . This yields a learned weighted shift $W_{\Xi_{sub}, c}$ and local interpolants $J_{0, \Xi_{sub}, c}, \dots, J_{K(\Xi_{sub}, c)-1, \Xi_{sub}, c}$. Define the orbit dictionary

$$\mathcal{D}(\Xi_{sub}, c) := \{J_{k, \Xi_{sub}, c} W_{\Xi_{sub}, c}^{(m)} x_i : (x_i, y_i) \in D_k, 0 \leq k \leq K(\Xi_{sub}, c) - 1, 0 \leq m \leq d-1\},$$

and the induced scaffold $U(\Xi_{sub}, c) := \text{span } \mathcal{D}(\Xi_{sub}, c) \subset \mathbb{R}^C$.

For each budget b , the primary geometric criterion is the certified projection error obtained by applying the scaffold-selection Algorithm ?? among all the scaffolds varying $\Xi_{sub} \in \mathfrak{X}_b$, and $c \in \mathcal{C}$ and get one selected scaffold $U_b^* := U(\Xi_{sub}^*(b), c^*(b))$. Let $f(x_i) = y_i$, for all $(x_i, y_i) \in \Xi_{train}$, be the target function, and define

$$\varepsilon_{proj}(b) := \frac{\sum_{(x_i, y_i) \in \Xi_{sub}^*(b)} \|f(x_i) - \Pi_{U_b^*} f(x_i)\|_2}{|\Xi_{sub}^*(b)|}, \quad \text{where } \Pi_U f(x_i) = UU^T f(x_i).$$

Let $\varepsilon_{min} := \min_{b \in \mathcal{B}} \varepsilon_{proj}(b)$ and given $\delta > 0$ form the budget δ -neighborhood $\mathcal{B}_\delta := \{b : \varepsilon_{proj}(b) \leq \varepsilon_{min} + \delta\}$.

This first stage is justified since in previous sections we identified scaffold geometry as a genuine bottleneck.

Under vanishing regime, the following decay estimate underlies an intrinsic denoising behaviour and controls the intrinsic memory horizon. Proof of Lemma 20 deferred to the Appendix A.

Lemma 20 (Noise Suppression via Orbit Decay) *Let $z \in \mathbb{R}^d$ and let $W^{(m)}$ with weights $w_0, \dots, w_{d-1} \in \mathbb{R}$ such that $\rho := \prod_{i=0}^{d-1} |w_i| < 1$. Then for all $m \geq 1$, $\|W^{(m)} z\| \leq \rho^{\lfloor m/d \rfloor} \cdot C$, for some constant $C > 0$ depending only on z and d . In particular, if z is an additive noise vector, then $\|W^{(m)} z\| \rightarrow 0$ exponentially as $m \rightarrow \infty$.*

Together, these effects give WBSNN a transparent mechanism for orbit stability, effective rank, and noise attenuation under the vanishing regime. More interpretable than standard neural architectures, whose contraction behavior has no comparable closed-form description.

Thus, Lemma 20 fully justifies that our next priority in selecting the final scaffold is vanishing regimes. If the weights of $W_{\Xi_{sub}^*(b), c^*(b)}$ are $w_0, \dots, w_{d-1}$, set $\rho(b) := \prod_{i=0}^{d-1} |w_i|$. Now define the vanishing-budget subset $\mathcal{B}_\delta^{van} := \{b \in \mathcal{B}_\delta : \rho(b) < 1\}$.

Case 1: If $|\mathcal{B}_\delta^{van}| = 1$, then b^* is the unique element of $\mathcal{B}_\delta$. The final scaffold is $U_{b^*}^*$. This rule gives strict priority to vanishing dynamics whenever projection quality is already essentially tied, which is consistent with the role of vanishing orbits as an implicit regularizer in WBSNN as proved in Lemma 20.

Case 2: If $|\mathcal{B}_\delta^{van}| > 1$, then among the budgets in $\mathcal{B}_\delta^{van}$ we select the scaffold minimizing a secondary score based on conditioning of the interpolants J_k , effective rank and compatibility with the vanishing regime as justified by Lemma 20.

So, the final selection rule should be $U_{b^*}^*$, where $b^* \in \arg \min_{b \in \mathcal{B}_\delta^{van}} \text{Score}_{van}(b)$, and

$$\text{Score}_{van}(b) := \omega_{van} \rho(b) + \omega_{cond} \kappa^{norm}(b) + \omega_{eff} r_{eff}^{norm}(b; \varepsilon),$$

with nonnegative weights summing to 1. Here:

(i) *Interpolation conditioning.* If $\mathcal{M}_k(\Xi_{\text{sub}}^*(b), c^*(b))$ is the local linear system used to build $J_{k, \Xi_{\text{sub}}^*(b), c^*(b)}$ on D_k (exactly or via a regularized/pseudoinverse solve in practice), define the interpolation conditioning of block D_k by $\kappa(\mathcal{M}_k) := \|\mathcal{M}_k\|_2 \|\mathcal{M}_k^\dagger\|_2$, if $\mathcal{M}_k$ has full column rank, otherwise set $\kappa(\mathcal{M}_k) = \infty$, where $\mathcal{M}_k^\dagger$ is the Moore-Penrose pseudoinverse of $\mathcal{M}_k$. So $\kappa(\mathcal{M}_k)$ measures how stable the local interpolation problem is on D_k , where smaller $\kappa(\mathcal{M}_k)$ means the local interpolation system is better conditioned, hence more numerically stable. The budget-level conditioning score is then

$$\kappa(b) := \max_{1 \leq k \leq K(\Xi_{\text{sub}}, c)} \kappa(\mathcal{M}_k(\Xi_{\text{sub}}^*(b), c^*(b))).$$

We then consider the normalized version: define $\kappa_{\min} := \min_{b \in \mathcal{B}_\delta^{\text{van}}}$, $\kappa_{\max} := \max_{b \in \mathcal{B}_\delta^{\text{van}}}$. If $\kappa_{\max} > \kappa_{\min}$, then set $\kappa^{\text{norm}}(b) := (\kappa(b) - \kappa_{\min}) / (\kappa_{\max} - \kappa_{\min})$, otherwise set $\kappa^{\text{norm}}(b) := 0$.

(ii) *Effective rank.* For a fixed tolerance $\varepsilon > 0$, let $M_{\text{eff}}(x_i; \varepsilon)$ be the effective horizon relative to x defined as the smallest m such that $\|W_{\Xi_{\text{sub}}^*(b), c^*(b)}^{(m)} J_{k, \Xi_{\text{sub}}^*(b), c^*(b)} x_i\| \leq \varepsilon$, with $(x_i, y_i) \in D_k$. Assume $\rho(b) < 1$, then by the decay estimate of Lemma 20 we have the explicit bound

$$M_{\text{eff}}(x; \varepsilon) < d \left\lceil \frac{\log(\varepsilon/C)}{\log \rho(b)} \right\rceil. \quad (14)$$

In the vanishing regime, this measures the number of geometrically active orbit directions before deeper iterates become negligible. Proof of (14) deferred to the Appendix A.

In particular, the *pointwise effective rank* defined by

$$r_{\text{eff}}(x_i; b, \varepsilon) := \dim \text{span} \{ W_{\Xi_{\text{sub}}^*(b), c^*(b)}^{(m)} J_{k, \Xi_{\text{sub}}^*(b), c^*(b)} x_i : (x_i, y_i) \in D_k, 0 \leq m < M_{\text{eff}}(x_i; \varepsilon) \},$$

is structurally bounded by the orbit dynamics. Finally, we define the *subset-level effective rank* by

$$r_{\text{eff}}(b; \varepsilon) := \frac{1}{|\Xi_{\text{sub}}|} \sum_{(x_i, y_i) \in \Xi_{\text{sub}}^*(b)} r_{\text{eff}}(x_i; b, \varepsilon),$$

In other words, $r_{\text{eff}}(b; \varepsilon)$ is the average number of geometrically active orbit directions before vanishing dynamics make deeper iterates negligible. So it measures intrinsic orbit complexity. We then consider the normalized version: set $r_{\min} := \min_{b \in \mathcal{B}_\delta^{\text{van}}} r_{\text{eff}}(b; \varepsilon)$ and $r_{\max} := \max_{b \in \mathcal{B}_\delta^{\text{van}}} r_{\text{eff}}(b; \varepsilon)$. If $r_{\max} > r_{\min}$, then set $r_{\text{eff}}^{\text{norm}}(b; \varepsilon) := (r_{\text{eff}}(b; \varepsilon) - r_{\min}) / (r_{\max} - r_{\min})$, otherwise set $r_{\text{eff}}^{\text{norm}}(b; \varepsilon) := 0$.

Thus, in the vanishing regime, smaller effective horizon means fewer active orbit directions, which is exactly the structural simplicity we want to reward together with interpolation conditioning once projection error is already essentially tied. We preferred normalization because it makes the weights interpretable since each term is in $[0, 1]$.

Case 3: If $|\mathcal{B}_\delta^{\text{van}}| = \emptyset$, then among the budgets in $\mathcal{B}_\delta$ we select the scaffold minimizing a penalized score $\text{Score}_{\text{novan}}(b)$. Thus, the final selection rule should be $U_{b^*}^*$, where $b^* \in \arg \min_{b \in \mathcal{B}_\delta} \text{Score}_{\text{novan}}(b)$, and

$$\text{Score}_{\text{novan}}(b) := \omega_{\text{novan}} V_{\text{novan}}^{\text{norm}}(b) + \omega_{\text{cond}} \kappa^{\text{norm}}(b),$$

where $V_{\text{novan}}(b) := \max\{0, \rho(b) - 1\}$, $V_{\min} := \min_{b \in \mathcal{B}_\delta} V_{\text{novan}}(b)$ and $V_{\max} := \max_{b \in \mathcal{B}_\delta} V_{\text{novan}}(b)$. If $V_{\max} > V_{\min}$, then define $V_{\text{novan}}^{\text{norm}}(b) := (V_{\text{novan}}(b) - V_{\min}) / (V_{\max} - V_{\min})$, otherwise set $V_{\text{novan}}^{\text{norm}}(b) := 0$. Thus, the non-vanishing penalty should increase as $\rho(b)$ moves away from the vanishing regime.

Phase 2: Let $f_{\text{CCDS}}(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} a_{k,m}^{\text{eff}}(x) u_{k,m}(x)$, with $(x, y) \in \Xi_{\text{train}}$, be the CCDS realizer on the optimal scaffold $U_{b^*}^*$ from Phase 1, where $\{u_{k,m}\}_{0 \leq k \leq K-1, 0 \leq m \leq d-1}$ is the family of innovation directions, with $u_{k,m} \neq 0$ for the indices selected by CCDS as defined in (??) and $u_{k,m} = 0$ for all remaining indices. Denote $a^{\text{eff}}(x) := (a_{k,m}^{\text{eff}}(x))_{k,m} \in \mathbb{R}^{Kd}$.

Phase 3: Mimicking dynamics in infinite dimensions, we define Phase 3 of WBSNN as a generalization mechanism over the orbit-induced class

$$\tilde{f}(x) = \sum_{k=0}^{K-1} \sum_{m \geq 0} \alpha_{k,m}(x) u_{k,m}(x), \quad x \in \mathbb{R}^d,$$

trained on the full dataset Ξ_{train} . Equivalently, Phase 3 learns over the family generated by the orbit geometry associated with all training samples, while producing a predictor defined for arbitrary inputs $x \in \mathbb{R}^d$. Here $\{u_{k,m}(x)\}_{k,m}$ denotes the family of CCDS innovation directions as defined in ??, extracted from the orbit atoms $q_{k,m}(x) = J_k W^{(m)} x$; in particular, these directions are obtained from the orbit family by orthogonalization against the previously selected scaffold directions, so the index m still retains its orbit-dynamical meaning. Note that this is the reason why we considered unwrapping $W^{(L)}$ modulo d ; otherwise, standard backward shifts would vanish beyond depth d , preventing generalization along the infinite orbit. By Lemma 3, in finite-dimensional spaces our generalization expression collapses—up to scalar factors to the prediction rule:

$$\tilde{f}(x) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}(x) \cdot u_{k,m}(x), \quad x \in \mathbb{R}^d. \quad (15)$$

According to the in-sample transfer Theorem 19, the Phase 3 predictor must be explicitly of the form prescribed by the prediction rule (15), and the associated coefficient map must approximate the effective CCDS coefficient map on that scaffold. Thus, Phase 3 is not trained to learn coefficients from scratch, but to approximate the constructive CCDS realization rule within the same orbit-based representation class.

The next step is to define the predictor $\tilde{f}$ explicitly so that: (i) it has the form prescribed by the prediction rule (15), as guaranteed by Proposition 22 below; (ii) it is endowed with an explicit guiding structure that allows it to track the CCDS coefficient map effectively; and (iii) it is implemented through a hybrid shallow policy, thereby preserving interpretability.

Definition 21 (Predictive Hybrid Learned CCDS mechanism of Phase 3) *Assuming the outputs of Phases 1 and 2 have already been obtained, we now introduce the hybrid Phase 3 mechanism, which is trained on the full dataset Ξ_{train} and yields a predictor defined on all of $\mathbb{R}^d$.*

Orbit dictionary: For each input $x \in \mathbb{R}^d$, define the orbit atoms

$$d_{k,m}(x) := J_k W^{(m)} x \in \mathbb{R}^C, \quad k \in \{0, \dots, K-1\}, \quad m \in \{0, \dots, d-1\},$$

and their normalized versions

$$\tilde{d}_{k,m}(x) := \frac{d_{k,m}(x)}{\|d_{k,m}(x)\| + \varepsilon_{\text{norm}}},$$

where $\varepsilon_{\text{norm}} > 0$ is fixed.

We write $\mathcal{D}(x) := \{\tilde{d}_{k,m}(x) : 0 \leq k \leq K-1, 0 \leq m \leq d-1\}$.

Learned policy variables: Let $T \geq 1$ be the number of Phase 3 steps. For each $t = 1, \dots, T$, let $g_\theta^{(t)} : \mathbb{R}^{d+C} \rightarrow \mathbb{R}^{Kd}$, $h_\theta^{(t)} : \mathbb{R}^{d+C} \rightarrow \mathbb{R}$, be shallow networks (e.g. one-hidden-layer MLPs). Their role is only to select the policy variables of the CCDS step: a soft atom-mixture and a feasible deviation slack. The CCDS coefficient itself remains explicit.

We also let $b_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be a shallow initialization map, used to define the residual target for the constructive recursion.

Initialization: Set $\hat{y}_0(x) := 0$, $r_0(x) := b_\theta(x)$, $Y_0(x) := \{0\}$.

Step $t = 1, \dots, T$. Assume $\hat{y}_{t-1}(x)$, $r_{t-1}(x)$, and $Y_{t-1}(x)$ are already defined. Define the state

$$z_t(x) := [x, r_{t-1}(x)] \in \mathbb{R}^{d+C}.$$

Soft policy over orbit atoms: Let $\pi_t(x) := \text{softmax}(g_\theta^{(t)}(z_t(x))) \in \Delta^{Kd-1}$, and define the soft selected direction $q_t(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \pi_{t;k,m}(x) \tilde{d}_{k,m}(x)$.

Innovation direction: Let $v_t(x) := q_t(x) - \Pi_{Y_{t-1}(x)} q_t(x)$. Assuming $\|v_t(x)\| \neq 0$, define

$$u_t(x) := \frac{v_t(x)}{\|v_t(x)\| + \varepsilon_{\text{orth}}}, \quad Y_t(x) := Y_{t-1}(x) \oplus \text{span}\{u_t(x)\},$$

where $\varepsilon_{\text{orth}} > 0$ is fixed.

Feasible deviation target: Set $a_t(x) := (I - \Pi_{Y_{t-1}(x)}) r_{t-1}(x)$, $\beta_t(x) := \langle a_t(x), u_t(x) \rangle$, and $d_{t,\min}(x) := \sqrt{\max(\|a_t(x)\|^2 - \beta_t(x)^2, 0)}$. Let $\Delta_t(x) := \text{softplus}(h_\theta^{(t)}(z_t(x))) \geq 0$, $d_t(x) := d_{t,\min}(x) + \Delta_t(x)$. Hence $d_t(x) \geq d_{t,\min}(x)$ by construction, so feasibility is automatic.

Explicit CCDS coefficient. Define the step coefficient by the CCDS root

$$\lambda_t(x) := \beta_t(x) - \sqrt{\max(d_t(x)^2 - d_{t,\min}(x)^2, 0)}.$$

(One may also choose the $+$ root if desired; throughout we fix one branch consistently.)

Residual update: Set $\hat{y}_t(x) := \hat{y}_{t-1}(x) + \lambda_t(x) u_t(x)$, $r_t(x) := b_\theta(x) - \hat{y}_t(x)$.

Final predictor: The Phase 3 output is $\tilde{f}(x) := \hat{y}_T(x)$.

To apply the in-sample transfer Theorem 19 to our predictor $\tilde{f}$ from Definition 21, it remains to verify that $\tilde{f}$ is of the form prescribed by (15), namely, that $\tilde{f}$ is given by an effective coefficient map over the innovation-direction family $\{u_{k,m}(x)\}_{k,m}$. Proposition 22 below provides precisely this result. So Phase 3 should be understood as a mechanism for constructing the effective coefficients $\alpha_{k,m}(x)$ in a structured and interpretable way, namely as an approximation of the CCDS coefficient map associated with the scaffold and innovation directions produced by Phases 1 and 2.

Proposition 22 (Hybrid CCDS Phase 3 has the required structural form) *Let $\tilde{f}$ be Phase 3 hybrid CCDS predictor from Definition 21, then*

- (i) *For each $t = 1, \dots, T$, there exists a coefficient vector $c_t(x) \in \mathbb{R}^{Kd}$ such that $u_t(x) = \tilde{D}(x) c_t(x)$.*
- (ii) *Consequently, for each x there exists an effective coefficient vector $\tilde{\alpha}^{\text{eff}}(x) := \sum_{t=1}^T \lambda_t(x) c_t(x) \in \mathbb{R}^{Kd}$ such that $\tilde{f}(x) := \hat{y}_T(x) = \tilde{D}(x) \tilde{\alpha}^{\text{eff}}(x)$.*
- (iii) *Defining $\alpha_{k,m}^{\text{eff}}(x) := \frac{\tilde{\alpha}_{k,m}^{\text{eff}}(x)}{\|d_{k,m}(x)\| + \varepsilon_{\text{norm}}}$, one has $\tilde{f}(x) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}^{\text{eff}}(x) J_k W^{(m)} x$.*

Proof We proceed by induction on t . For $t = 1$, we have $Y_0(x) = \{0\}$, hence $v_1(x) = q_1(x) = \tilde{D}(x) \pi_1(x)$. Since $v_1(x) \neq 0$, $u_1(x) = \frac{v_1(x)}{\|v_1(x)\|} = \tilde{D}(x) \frac{\pi_1(x)}{\|\tilde{D}(x) \pi_1(x)\|}$. Thus (i) holds for $t = 1$ with $c_1(x) := \frac{\pi_1(x)}{\|\tilde{D}(x) \pi_1(x)\|}$.

Assume now that for some $t \geq 2$, vectors $c_1(x), \dots, c_{t-1}(x)$ have already been constructed so that $u_s(x) = \tilde{D}(x) c_s(x)$, $s = 1, \dots, t-1$. Because $u_1(x), \dots, u_{t-1}(x)$ are orthonormal by construction,

$$\Pi_{Y_{t-1}(x)} q_t(x) = \sum_{s=1}^{t-1} \langle q_t(x), u_s(x) \rangle u_s(x).$$

Since $q_t(x) = \tilde{D}(x) \pi_t(x)$ and $u_s(x) = \tilde{D}(x) c_s(x)$, it follows that

$$\Pi_{Y_{t-1}(x)} q_t(x) = \tilde{D}(x) \sum_{s=1}^{t-1} \langle q_t(x), u_s(x) \rangle c_s(x).$$

Hence $v_t(x) = q_t(x) - \Pi_{Y_{t-1}(x)} q_t(x) = \tilde{D}(x) \nu_t(x)$, where $\nu_t(x) := \pi_t(x) - \sum_{s=1}^{t-1} \langle q_t(x), u_s(x) \rangle c_s(x)$. Since $v_t(x) \neq 0$, we have that $u_t(x) = \frac{v_t(x)}{\|v_t(x)\|} = \tilde{D}(x) \frac{\nu_t(x)}{\|\tilde{D}(x) \nu_t(x)\|}$. Thus (i) holds for step t with $c_t(x) := \frac{\nu_t(x)}{\|\tilde{D}(x) \nu_t(x)\|}$. This completes the induction and proves (i).

For (ii), using (i) and the update rule,

$$\hat{y}_T(x) = \sum_{t=1}^T \lambda_t(x) u_t(x) = \sum_{t=1}^T \lambda_t(x) \tilde{D}(x) c_t(x) = \tilde{D}(x) \sum_{t=1}^T \lambda_t(x) c_t(x).$$

Thus $\tilde{f}(x) = \tilde{D}(x) \tilde{\alpha}^{\text{eff}}(x)$ with $\tilde{\alpha}^{\text{eff}}(x) := \sum_{t=1}^T \lambda_t(x) c_t(x)$.

For (iii), by the definition of the normalized atoms, $\tilde{d}_{k,m}(x) = \frac{d_{k,m}(x)}{\|d_{k,m}(x)\| + \varepsilon_{\text{norm}}}$, so the representation with respect to $\tilde{D}(x)$ is equivalent to a representation with respect to the original orbit atoms $d_{k,m}(x) = J_k W^{(m)} x$, obtained by dividing the effective normalized coefficients by the same normalization factors. Therefore $\tilde{f}(x) = \sum_{k,m} \alpha_{k,m}^{\text{eff}}(x) J_k W^{(m)} x$. This completes our proof. $\blacksquare$

To enforce both prediction accuracy and coefficient tracking, we train Phase 3 by combining the usual prediction loss with a coefficient-supervision loss against the effective CCDS coefficients. The resulting Phase 3 loss is

$$\mathcal{L}_{\text{Phase 3}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{coeff}} \frac{1}{n} \sum_{i=1}^n \|\alpha(x_i) - \alpha^{\text{eff}}(x_i)\|_{\ell^2}.$$

Thus, the in-sample predictive bound of Theorem 19 becomes exactly:

$$\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \underbrace{\|f - \Pi_{U_{b^*}^*} f\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{proj}}(f, U_{b^*}^*, \Xi_{\text{train}})} + \underbrace{\|\Pi_{U_{b^*}^*} f - f_{\text{CCDS}}\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{param}}(f_{\text{CCDS}}, U_{b^*}^*, \Xi_{\text{train}})} + M_u \frac{1}{n} \sum_{i=1}^n \|\alpha^{\text{eff}}(x_i) - \alpha(x_i)\|_{\ell^1}, \quad (16)$$

where $f(x_i) = y_i$ for all $(x_i, y_i) \in \Xi_{\text{train}}$; $M_u = \max_{0 \leq k \leq K-1, 0 \leq m \leq d-1, 1 \leq i \leq n} \|u_{k,m}(x_i)\|_{\ell^2}$; $a^{\text{eff}}(x_i) = (a_{k,m}^{\text{eff}}(x_i))_{k,m} \in \mathbb{R}^{Kd}$ and $\alpha(x_i) = (\alpha_{k,m}(x_i))_{k,m} \in \mathbb{R}^{Kd}$.

Hence, in-sample prediction succeeds when the scaffold selected in Phase 1 is good enough that the projected target is close to the true dataset map, the Phase 2 CCDS realizer on that scaffold is good enough to provide the model with a structured surrogate of the projected map, so that the target map is not learned in an unconstrained way, and finally the hybrid Phase 3 policy is good enough to track the CCDS coefficient map.

Remark 23 *The reader should note that:*

- (i) *If the Phase 2 CCDS realizer collapses to the degenerate MP regime, then the CCDS innovation directions reduce to the raw orbit atoms $J_k W^{(m)} x$, and the prediction rule (15) becomes*

$$\tilde{f}(x) = \sum_{k=0}^{K-1} J_k \sum_{m=0}^{d-1} \alpha_{k,m}(x) W^{(m)} x, \quad x \in \mathbb{R}^d.$$

- (ii) *Phase 3 is deliberately defined on infinite orbits in order to pave the way for the extension of WBSNNs to infinite-dimensional spaces. This is fully natural in our framework, since both the exact interpolation theorem and the CCDS mechanism hold in infinite-dimensional Banach spaces. In this paper, however, we restrict our attention to the finite-dimensional case; a full treatment of the infinite-dimensional WBSNN framework is left for future work.*

We are now ready to formally introduce WBSNNs in the finite-dimensional setting.

Definition 24 (WBSNN on a finite-dimensional dataset) *Given a finite-dimensional training dataset Ξ_{train} , a Weighted Backward Shift Neural Network (WBSNN) associated to Ξ_{train} is the predictor $\tilde{f}$ obtained from the three-phase pipeline of this paper: Phase 1 selects a scaffold U_{b^*} , Phase 2 constructs on that scaffold a CCDS realizer of the projected target, and Phase 3 learns a data-dependent coefficient map $\alpha_{k,m}(x)$ through the hybrid predictive CCDS mechanism from Definition 21, yielding the final prediction rule $\tilde{f}$ via (15).*

Moreover, the predictor $\tilde{f}$ from Definition 24 satisfies on Ξ_{train} the in-sample prediction error bound given in (16).

Remark 25 *WBSNN is a general-purpose learning paradigm that operates across modalities and tasks without requiring task-specific architectural redesign. It replaces stacked nonlinearities with structured orbit dynamics. Moreover, Theorem 19 is only an in-sample structural result: it does not by itself establish test generalization. That question is addressed later in Theorem 26. Nevertheless, the framework already suggests a structural inductive bias favorable to robust generalization: optimal scaffold budgets limit capacity, the orbit-span constraint restricts the hypothesis class, vanishing dynamics suppress noise, relaxed interpolation improves stability in noisy regimes, and the shallow Phase 3 head only reweights orbit atoms rather than learning an unrestricted predictor.*

Under the additional assumption that test points lie near the same orbit geometry captured at training, we obtain an out-of-sample prediction bound that preserves the same structural logic as the in-sample bound from Theorem 19. Thus, although the WBSNN architecture already exhibits a structural inductive bias favorable to generalization, an explicit out-of-sample result is still needed to complete the predictive picture of the framework.

see if we need any of this preamble, we have already defined test set at the very beginning of the section Let $U \subset \mathbb{R}^C$ be the scaffold selected from Ξ_{train} , and let $f : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be the target map, so that $f(x_i) = y_i$, $f(\bar{x}_j) = \bar{y}_j$. Let $\tilde{f}$ be the predictor built on the scaffold U .

the lipchitz conditions in next thm are not necessary to ask as they can be proved as part of the theorem that they hold, make this change in the theorem

Theorem 26 (Geometric out-of-sample prediction bound) *Assume*

1. (Neighborhood relation between test and train) *There exists $\delta \geq 0$ such that for every $\bar{x}_j \in \Xi_{test}$ there exists some $x_{i(j)} \in \Xi_{train}$ with $\|\bar{x}_j - x_{i(j)}\|_2 \leq \delta$.*
2. (Regularity of the projected target) *The map $\Pi_U f$ is L_{Π_U} -Lipschitz on the δ -neighborhood of Ξ_{train} : $\|\Pi_U f(x) - \Pi_U f(x')\|_2 \leq L_{\Pi_U} \|x - x'\|_\ell^2$, $\forall x, x' \in \Xi_{train} \cup \Xi_{test}$.*
3. (Regularity of the predictor) *The map $\tilde{f}$ is $L_{\tilde{f}}$ -Lipschitz on the same neighborhood: $\|\tilde{f}(x) - \tilde{f}(x')\|_2 \leq L_{\tilde{f}} \|x - x'\|_\ell^2$, $\forall x, x' \in \Xi_{train} \cup \Xi_{test}$.*

Then

$$\|f - \tilde{f}\|_{\Xi_{test}} \leq \underbrace{\|f - \Pi_U f\|_{\Xi_{test}}}_{\varepsilon_{proj}(f, U, \Xi_{test})} + \max_{x: (x, y) \in \Xi_{train}} \|\Pi_U f x - \tilde{f} x\|_{\ell^2} + (L_{\Pi_U} + L_{\tilde{f}})\delta.$$

Proof Fix $j \in \{1, \dots, m\}$, and choose $x_{i(j)} \in \Xi_{train}$ such that $\|\bar{x}_j - x_{i(j)}\|_{\ell^2} \leq \delta$. Then

$$f(\bar{x}_j) - \tilde{f}(\bar{x}_j) = (f(\bar{x}_j) - \Pi_U f(\bar{x}_j)) + (\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})) + (\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})) + (\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)).$$

Applying the triangle inequality,

$$\begin{aligned} \|f(\bar{x}_j) - \tilde{f}(\bar{x}_j)\|_2 &\leq \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_2 + \|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_2 \\ &\quad + \|\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})\|_2 + \|\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)\|_2. \end{aligned}$$

By Assumption 2, $\|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_2 \leq L_{\Pi_U} \delta$. By Assumption 3, $\|\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)\|_{\ell^2} \leq L_{\tilde{f}} \delta$. Therefore,

$$\|f(\bar{x}_j) - \tilde{f}(\bar{x}_j)\|_{\ell^2} \leq \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_{\ell^2} + \|\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})\|_{\ell^2} + (L_{\Pi_U} + L_{\tilde{f}}) \delta.$$

Averaging over $j = 1, \dots, m$ we obtain

$$\|f - \tilde{f}\|_{\Xi_{test}} \leq \|f - \Pi_U f\|_{\Xi_{test}} + \max_{x:(x,y) \in \Xi_{train}} \|\Pi_U f x - \tilde{f} x\|_{\ell^2} + (L_{\Pi_U} + L_{\tilde{f}}) \delta.$$

This proves the theorem. ■

The out-of-sample version preserves the same logic as the in-sample version: (i) $\|f - \Pi_U f\|_{\Xi_{test}}$ is the scaffold bottleneck on test; (ii) $\max_{x:(x,y) \in \Xi_{train}} \|\Pi_U f(x) - \tilde{f}(x)\|_{\ell^2}$ is the predictor bottleneck inherited from training; and (iii) $(L_{\Pi_U} + L_{\tilde{f}}) \delta$ is the new transfer term, which says that test points must lie near the training scaffold geometry. If $\delta = 0$, Theorem 26 collapses back to the same two bottlenecks: projection error on test and parametrization error of the predictor on training.

Visualizing WBSNN in Action: Interpretability on Swiss Roll. split it in interpretability per phases 1 to 3 with new phase 3 To illustrate WBSNN’s interpretability and operational mechanics, we examine its application to the Swiss Roll dataset—a classic nonlinear manifold benchmark—with $d = 3$, 120 samples, and added Gaussian noise of 0.5. In this run, Phase 1 and Phase 2 use approximately 6% of the training set (~ 5 points) for learning the shift operator W and subsets D_k for interpolation, while Phase 3 leverages the full training data for generalization. Critical numerical details appear in Appendix ??, Table 1.

Phase 1 begins by selecting a subset of training points and optimizing $W = [0.8, 0.8, 0.8]$, partitioning them into 2 subsets D_0 and D_1 (total 5 points) with $\Delta = 3.1146$. In Phase 2, local operators J_0 and J_1 are constructed to achieve exact interpolation (norm differences ≈ 0 within numerical precision, e.g., 7.77×10^{-16} for Point 1 under J_0), confirming Theorem 2.7’s guarantees. This exact fit on subsets ensures the model captures local geometry without overfitting the full dataset. The final performance, shown in Table 1, positions WBSNN as data-efficient and robust: achieving a test accuracy of 0.9583 and train accuracy of 0.9868, competitive with baselines despite using far fewer samples in its core phases. This extreme data efficiency (6% vs. 100% for baselines) underscores WBSNN’s ability to capture manifold structure with minimal data. Unlike baselines, which may memorize the small dataset, WBSNN leverages orbit dynamics for robust generalization. Phase 3’s MLP outputs $\alpha_{k,m}$ weights, which assign explicit contributions to each projection and dimension, enabling direct inspection of decision factors unlike black-box baselines, enabling traceable predictions via Equation (15), i.e. $\hat{Y} = \sum_{k=0}^1 J_k \sum_{m=0}^2 \alpha_{k,m}(X) \cdot W^{(m)} X$.

To visualize WBSNN’s interpretability, we examine predictions for two points: (1) a held-out training point not used in Phase 1 (seen by the model during Phase 3 training but excluded from subset construction in Phases 1–2), and (2) an unseen test point (never encountered during any phase). This selection highlights WBSNN’s generalization: the held-out training point tests interpolation beyond Phase 1 subsets (demonstrating robustness within the training distribution), while the test point evaluates true out-of-sample performance (showing adaptability to unseen manifold regions). Both points yield correct predictions, underscoring the framework’s ability to leverage orbit dynamics for reliable, traceable decisions.

For the held-out training point, the prediction sum from Phase 3 gives (2.328883) yielding a sigmoid probability of 0.911241, confidently predicting label 1.0 (matching true). The high probability

(0.911241) reflects strong confidence in Class 1, driven by positive terms $\alpha_{k,m}$ from dimensions 1 and 2, aligning with the spiral’s geometry. For the unseen test point, the sum (-2.475847) yields a sigmoid probability of 0.077569, predicting label 0.0 (matching true). The low probability (0.077569) indicates robust separation from Class 1, with negative $\alpha_{k,m}$ suppressing projections across all dimensions, illustrating WBSNN’s ability to generalize to unseen points by adapting weights to the input’s position on the manifold.

These examples, detailed in Appendix ??, Table 1, highlight WBSNN’s transparency: each prediction decomposes into per-dimension contributions. The selection of these points underscores WBSNN’s strengths: the held-out training point shows how the model interpolates within the training distribution without relying on Phase 1 subsets, while the test point validates true extrapolation. This dual view emphasizes interpretability—each prediction traces back to explicit orbit combinations—positioning WBSNN as a transparent alternative to black-box models.

Add explanation about interpretability of hybrid Phase3: this explanation that follows is not accurate, update and give numbers....Each block mirrors a CAN/CCDS step: Activations are inspectable residuals, and contributions trace back to specific orbit atoms. Unlike a generic MLP, this does not add nonlinear expressivity—the scaffold span bounds the class, with learning optimizing the solver’s pursuit path for generalization.

On $\alpha_{k,m}$ modalities. Concerning the MLP-learned coefficient $\alpha_{k,m}$ of our prediction rule (15), we consider four WBSNN modalities along two axes—head expressivity and backbone sharing. **(1) $\alpha_{k,m}$:** a full per- k , per-state head that gates all dynamical states m within each subset k (maximal flexibility). **(2) $\alpha_{k,L}$:** a middle-ground head that gates only the states established by Phase 1 and 2, meaning for each $0 \leq k \leq K - 1$, the coefficient $\alpha_{k,L} = 0$ for all $L \notin \mathcal{L}_k$, where $\mathcal{L}_k$ is the set of L_i satisfying condition (3) for $X_i \in D_k$, reducing variance while retaining within- k selectivity. **(3) α_k (shared backbone):** a per- k head trained on a trunk *primed by $\alpha_{k,m}$* ; operationally it aggregates over m using features learned for the richer head, offering strong generalization with lower overfitting. **(4) α_k (separate backbones):** the same per- k head trained on its own trunk, decoupled from $\alpha_{k,m}$. So, $\alpha_{k,L}$ is an intermediate head that sits between $\alpha_{k,m}$ and α_k . Our main protocol uses variants (1) and (3), i.e. full $\alpha_{k,m}$ and α_k with a shared, $\alpha_{k,m}$ -primed backbone, balancing accuracy, robustness, and efficiency. Appendix ?? reports the behavior of $\alpha_{k,m}$ and both α_k variants across datasets. We reserve the variant $\alpha_{k,L}$ exclusively for the infinite-dimensional formulation of WBSNNs (Appendix ??), where it plays a central role.

Interpolation modalities. already said if we paste realizability paper before this where we need it since we use relaxed int in realizability experiemnts, see if we can enrich the text tehre While Theorem 4 provides a constructive proof of exact interpolation via sequential updates of the operator J , our implementation adheres to the same design principles while allowing a relaxed (non-exact) interpolation adapting to realistic settings for stability, efficiency, and robustness. This last variant intentionally relaxes the exact fit to illustrate the model’s behavior under noisy conditions, offering a broader view of orbit-based generalization and adaptive weighting phenomena. Details are deferred to Appendix B. This dual approach allows us to adopt exact interpolation in noiseless or low-noise regimes and to switch to non-exact formulations when data is noisy or ill-conditioned.

Practical considerations. Phase 1 evaluates $W^{(L)}X$ across depths $L \leq d$ and tests span membership via a least-squares residual with tolerance τ (a numerically cheap proxy for exact independence; see Appendix B); the work grows roughly quadratically with the subset size. To keep costs tractable without losing diversity, we optimize W on mini-subsets and form D_k from a subsampled fraction of the training pool (diversity-aware anchor policies—e.g., coverage/ k -center ideas—instead of uniform subsampling are left for future work). In Phase 2 we compute J_k using either the constructive rank-one update scheme given by the proof of Theorem 4 (yielding exact interpolation), the closed-form pseudoinverse (still giving exact interpolation when full-rank), or a regularized pseudoinverse for noisy/ill-conditioned cases yielding relaxed interpolation, details in (Appendix B). Our main results default to the regularized form for stability in noisy/compressed regimes, with exact variants used when independence is clean. Crucially, we handle vector and

scalar targets uniformly in the same d -dimensional space: classification uses one-hot label vectors, and regression embeds the scalar along a fixed direction and reads it back through a fixed linear functional. Phase 3 then reuses the same orbit-weighted combination, preserving interpretability.

Empirical Observation (implicit regularization). Across nine datasets and varied configurations of input dimension and sample size, the learned shift W almost always lies in the vanishing regime, $\prod_{i=0}^{d-1} w_i < 1$. This contraction behaves as an implicit regularizer—improving generalization and noise robustness—while rare cases favor neutral or mildly expanding dynamics to preserve signal, reflecting a regularization–expressivity trade-off. This motivates the next lemma.

Noise Suppression via Orbit Decay. add the effective rank result in a unified way WBSNN generalizes over the infinite orbit $\{W^{(m)}X\}_{m \in \mathbb{N}}$, which, by Lemma 3, collapses—up to scalars—to its first d iterates. In the vanishing regime—where $\rho := \prod_{i=0}^{d-1} |w_i| < 1$ —each $W^{(m)}\varepsilon$ decays as $\rho^{\lfloor m/d \rfloor}$, producing spectral regularization. Though Phase 3 only uses shallow iterates, the full decay structure biases the model toward stability. This yields noise suppression without explicit penalties (like dropout layers), driven solely by the orbit geometry of the shift operator itself. The result is stable learning behavior in noisy or low-sample settings, as formally shown below.

Remark 27 *This shows that although Phase 3 in WBSNN only uses orbit powers $m \leq d - 1$, these terms implicitly encode decay across the full infinite orbit $\{W^{(m)}\varepsilon\}_{m \geq 0}$ via scalar rescaling. The vanishing regime $\prod_{i=0}^{d-1} w_i < 1$ induces exponential suppression of long-term perturbations, acting as an intrinsic regularizer in the orbit geometry.*

For a concrete instantiation of our main protocol: relaxed (non-exact interpolation), see Algorithm ?? in Appendix ??.

3.2 WBSNN in an Operator-Theoretic Machine Learning Context.

Operator theory has shaped machine learning in diverse ways, primarily through Koopman operator frameworks, which linearize nonlinear dynamical systems via spectral decompositions (often approximated in practice via DMD or neural networks) for tasks like prediction and control (13), (11), (5). Extensions such as Koopman autoencoders further embed states into linear subspaces for generative modeling (1). Another class of models, such as DeepONet (10) and neural integral equations (19), learns mappings between infinite-dimensional function spaces, often for PDEs, using architectures like branch–trunk networks for DeepONet or attention-based solvers for neural integral equations. In contrast, models like Linformer (18) proposing a low-rank projection to reduce self-attention complexity from quadratic to linear, while NAIS-Net (6) enforces stability through non-autonomous differential equation dynamics. Collectively, these methods prioritize universal operator approximation or architectural constraints for efficiency and stability, yet they are not orbit-based and thus do not build representations through iterates of dynamic operators. These models serve as baselines in our experiments, against which WBSNN demonstrates superior data efficiency and noise robustness across diverse tasks.

In contrast, WBSNNs are not based on approximating nonlinear operators or constraining spectra for stability. Instead, they are rooted directly in dynamics of linear operators: the study of how iterates of a single operator generate rich geometric and topological behavior. A central object here is the hypercyclic operator—one admitting a vector whose orbit under iteration is dense in the space—with the weighted backward shift as the archetypal example, see (3). At first glance this notion seems exotic, since proving hypercyclicity often requires delicate constructions. Yet the key discovery is that hypercyclicity is not exceptional but typical: once one dense orbit exists, many do—and even more surprising, the set of hypercyclic operators living on any infinite-dimensional separable Banach space is itself topologically large, see Theorem 1.2, Proposition 2.20 and Corollary 2.9 (3). Motivated by these striking and unexpected facts, linear dynamics has become a crossroads where seemingly unrelated areas of mathematics converge, revealing a rich source of insights. WBSNNs borrow this spirit, grounding representation learning in orbit dynamics to bring interpretability

ity and expressive power into machine learning. By harnessing linear dynamics, WBSNN achieves data-efficient, noise-robust generalization—often with as low as 1% of training data—while maintaining exact interpolation and topological awareness, outperforming baselines in diverse regimes from manifolds to noisy signals.

3.3 Experiments

Protocol and compute. We report results across nine datasets under one protocol: train-only standardization, small PCA bottlenecks, and tiny Phase-1/2 “discovery” budgets to build the orbit dictionary. All experiments are conducted in finite-dimensional settings, and in every case the orbit scaffold is constructed from randomly selected anchors rather than via any optimized selection procedure; despite this minimal choice, WBSNN consistently achieves strong performance across datasets. Unless noted otherwise, α_k denotes the shared-trunk head (the main protocol), and $\alpha_{k,m}$ is the richer head gating per subset and per orbit state. All experiments run in a lightweight CPU-only Jupyter environment—no GPUs—so the numbers reflect true data/geometry leverage rather than horsepower.

Baselines. We report tuned baselines spanning: linear (LogReg, LinearSVM), kernel (RBF-SVM, KRR, Nyström, RFF), tree ensembles (RF, XGBoost, ExtraTrees), neural (MLP, CNN/LeNet/ResNet-18), sequence/dynamical (LSTM, Linformer, NAIS-Net), operator-learning (DeepONet, FNO), spectral/graph and nonparametric (Diffusion Maps+LogReg, Laplacian-LogReg, Scattering2D, Hankel-SVD+LDA, k -NN/Label Propagation), and domain-specific (CORAL, time-delay ridge, EDMD).

CIFAR-100 (compressed, capped). On a small-but-stubborn CIFAR-100 variant (30 classes, frozen ResNet-18 features, PCA to $d \in \{10, 20\}$ and 1–3% discovery), compressing to $d=10$ with a 3k budget keeps only $\sim 31\%$ variance and creates a low-margin regime: linear and kernel heads hover around 50%, while $\alpha_{k,m}$ sits between 45–47% and outperforms α_k at this extreme compression. Widening to $d=20$ ($\sim 43\%$ variance) and a 5k budget pushes $\alpha_{k,m}$ into the high-50s, narrowing the gap to the best shallow baselines despite discovery still capped at 3%—evidence that widening the bottleneck and modestly enlarging the pool helps the orbit dictionary organize what signal survives projection, see Tables ?? and ?? in the Appendix for details.

Gas Sensor Drift (chronological, distribution shift). The chronology (10 batches over 36 months; strict out-of-time split; PCA $d \in \{5, 10, 15\}$; discovery 1–7%) stresses noise and drift simultaneously. Here, WBSNN’s stability under drift and noise shone through. At $d=5$ with $n \approx 1.6k$ and 5–7% discovery, all WBSNN heads land ~ 59 – 61% with α_k (separate) 7% discovery outperforming the strongest baseline overall under drift. Opening to $d=10$ and $n=3k$ (EVR=96%) reveals the scaling law cleanly: $\alpha_{k,m}$ peaks near 68% at 3% discovery and remains best overall; both α_k variants are more budget-hungry but converge within a point or two by 5%. A 20-seed sweep at $d=10$ (10% discovery) shows $\alpha_{k,m}$ 0.647 ± 0.0367 , α_k (shared) 0.645 ± 0.051 , α_k (separate) 0.65 ± 0.037 —modest spread under drift and reproducible gains from sparse discovery. More details in Tables ??–??.

IMDb (compressed text, mean-pooled). With mean-pooled GloVe-100 embeddings, PCA $d \in \{10, 20, 50\}$, and 1–15% discovery, WBSNN scales predictably despite erasing word order. With 2k samples at $d=10$ (EVR ≈ 0.52), $\alpha_{k,m}$ holds near 0.75 (best overall); with 4k at $d=20$ (EVR ≈ 0.67) it rises to ~ 0.77 (best overall with Random Forest); with 25k at $d=50$ (EVR ≈ 0.87) and just 1% discovery it reaches ~ 0.793 , keeping pace with RBF-SVM/Nyström and outrunning logistic/GBDT. Exact-interpolation runs confirm the bias–variance trade: at low data they match or beat non-exact (e.g., ~ 0.73 – 0.77 at $d \in \{10, 20\}$ with 2–4k) with zero anchor residuals, while non-exact becomes preferable as samples and horizon grow. More details in Tables ??–??.

MNIST (PCA bottlenecks). Compressed regimes expose structure vs. capacity. At $d=5$ with 2k samples ($\sim 33\%$ variance), accuracy is limited by information loss; WBSNN remains competitive with SVM and consistently beats logistic/MLP/forest. At $d=15$ with 5k samples ($\sim 58\%$ variance), both heads climb; $\alpha_{k,m}$ benefits more from the extra signal. At $d=30$ with 10k samples ($\sim 73\%$ variance), WBSNN exceeds 95% test accuracy—rivaling strong PCA baselines (in both $d = 15, d = 30$)

without image-specific machinery—while accuracy rises monotonically with Phase-budget. Details in Tables ??-??.

Noisy synthetic gauntlet (heavy tails, spurious cue). A $15k \times 50D$ task with few informative coordinates, heavy-tailed heteroskedastic noise, boundary-peaked label flips, and a 0.6-correlated spurious feature separates information-seekers from shortcut-takers. Under tight discovery budgets, WBSNN recovers signal reliably: at small d it achieves exact interpolation (Phase-2 residuals ≈ 0) yet generalizes; as d ($5 \rightarrow 20$) or n ($1.5k \rightarrow 12k$) grow, accuracy rises to ≈ 0.75 – 0.78 , matching or slightly edging strong tabular/kernel baselines (LogReg, RBF-SVM, RFF+LogReg, ExtraTrees) and a tuned 1-layer MLP. Random forests overfit, while WBSNN remains stable and spurious-resistant, with predictable gains from modest increases in d or n . More details in Tables ??-??.

FI-2010 (limit order book). Chronological splits; PCA $d \in \{10, 20, 40\}$; discovery 1–10%; $n = 1k$ – $30k$. In the harsh corner $d=10, n=1k$, $\alpha_{k,m} \sim 0.458$ and α_k $0.404 \rightarrow 0.476$ (as budget rises) sit with LR/RBF-SVM/Nyström/MLP/RF around 0.40–0.47; Phase-2 already aligns most anchors exactly. At $d=20, n=2k$, both heads firm near 0.49 (best overall), ahead of Linformer/NAIS-Net/DeepONet/LR/MLP. Push to $d=20, n=10k$: WBSNN jumps to ~ 0.62 ; at $d=20, n=30k$ with just 1–3% discovery, both heads reach ~ 0.65 while baselines lag in the 0.44–0.59 range. A 20-seed small-data sweep ($d=10, n=2k$, 10%) yields 0.504 ± 0.046 for $\alpha_{k,m}$ and 0.500 ± 0.051 for α_k : typical low- n dispersion that tightens as information grows. Details in Tables ??-??.

PRSA-2017 (air-quality regression). Four meteorological features; strict chronological split where the last 20% ($\sim 82k$ hours) is the fixed test window. WBSNN’s $\alpha_{k,m}$ scales from $R^2 \approx 0.27$ at $n=1k$ to 0.29–0.30 at 3k and 0.32–0.33 at 10k with discovery ranging 1–15%, while MAE stays ~ 0.58 – 0.60 . The head hierarchy is consistent: $\alpha_{k,m}$ leads; α_k (shared) tracks closely and beats α_k (separate) as data grow. Exact vs. non-exact shows the intended trade-off: exact wins at $n=1k$ (e.g., $R^2 \approx 0.283$ – 0.284 vs. ~ 0.27) and stays similar at 3k, but non-exact pulls ahead at 10k (~ 0.326 – 0.332 vs. ~ 0.297 – 0.320), suggesting a small slack in Phase-2 improves drift-robustness on a massive future window. Proportional 80/20 ablations (e.g., 3k/600 and 10k/2k) mirror this: exact is stable on small tests; non-exact scales better on broader-horizon evaluations. Tables ??-??.

ISOLET (speech; 26-way). A clean compression/budget sweep. In the compressed corner ($d=5, n=2000$), $\alpha_{k,m}$ climbs from 62.0% $\rightarrow$ 64.8% as discovery rises 5% $\rightarrow$ 15%, trailing only RBF-SVM and beating logistic/MLP/RF/k-NN/Nyström/Label-Prop; α_k (shared) sits just behind and clearly outperforms α_k (separate). A 20-seed study at $d=5$, 10% shows $\alpha_{k,m}$ 0.62 ± 0.011 , α_k (shared) 0.58 ± 0.019 , α_k (separate) 0.55 ± 0.022 —sharing the trunk lowers variance. At $d=10, n=4000$, $\alpha_{k,m}$ reaches 77.1% (5%), essentially tied with RBF-SVM (77.6%) and ahead of other baselines. On the full train, tight $d=5$ caps at $\sim 65\%$, while $d=20$ yields $\sim 90\%$ and $d=35$ 91.9–92.4% with just 3–5% discovery—competitive with logistic/MLP and within a few points of RBF-SVM, all while forests/k-NN overfit. More details in Tables ??-??.

Swiss Roll + RFF (classification & regression). Three tough classification tracks and a heteroskedastic regression confirm that tiny discovery budgets suffice when geometry is organized. In noisy 5-class, small runs (RFF20 $\rightarrow$ PCA10, $M_{\text{train}}=800$, 15%) already hit $\sim 95\%$ ($\alpha_{k,m}$) in line with RBF-SVM; scaling to RFF30 $\rightarrow$ PCA15 with $M_{\text{train}}=30k$ and just 7% anchors keeps both heads $\sim 96\%$. In low-sample + label-noise ($\sim 20\%$ flips), the simpler α_k (shared) can edge $\alpha_{k,m}$ at higher d and M (e.g., $\sim 75\%$ at RFF20 $\rightarrow$ PCA15, 1k, 10%), consistent with a helpful bias under uniform flip noise. On multi-roll (4 spirals), WBSNN rides the topology: ~ 94 – 95% at RFF20 $\rightarrow$ PCA10, 2k/10% and $\sim 97\%$ at RFF30 $\rightarrow$ PCA15, 20k/7%, neck-and-neck with the best kernels/MLPs. For regression, α_k is the safer bias-controlled choice at scale (e.g., $R^2 \approx 0.83$ at 20k vs. ~ 0.78 for $\alpha_{k,m}$), showing that wider heads add variance when signal is essentially 1-D. Further details in Tables ??-??.

Swiss Roll + polynomial embedding. With explicit quadratics + interactions + six spurious channels, WBSNN behaves like a well-regularized model. At mild noise (0.2), it tracks 98.1% and logistic wins by a hair 98.2%—exactly what you expect in a quasi-linearized space. At noise 0.5, WBSNN holds mid-95s, with smaller d acting as a built-in denoiser (e.g., 95.3% at $d=10, 5k$ vs. 93.6% at $d=15, 5k$); at noise 0.8, small- d plus small budgets remains the safe recipe, and WBSNN

stays competitive while kernel/linear baselines occasionally nip it by 0.5–1 point—again consistent with bias–variance under noisy algebraic features. More details in Tables ??–??.

Swiss Roll (raw 3D). No feature maps, just standardization—WBSNN treats geometry as home turf. With three noisy classes ($\sigma=0.5$) and only $\sim 10\%$ discovery, it is already 99.4% at 10k. Logistic stalls at 61%, underscoring that this is about geometry, not head capacity. Under scarcity + label noise (10-way, 10% flips, 1k points), WBSNN holds 91.5% and matches or edges kernels. On the 3-roll topology at 20k, it again rides with the leaders ($\sim 98.7\%$) with only 7% budget. For regression on the unwrapped angle, the shared α_k head ages better as scale increases (e.g., $R^2 \approx 0.986$ at 2k vs. ~ 0.969 for $\alpha_{k,m}$, and $R^2 \approx 0.961$ at 10k vs. ~ 0.938), highlighting that a narrower head can be the right bias when the target is essentially 1-D. Further details in Tables ??–??.

Conclusion. Across a deliberately eclectic suite of datasets—under stressors including heavy/heteroskedastic noise, distribution drift, extreme PCA compression, label flips, spurious cues, and multi-roll topology—the pattern is consistent. With tiny discovery budgets (often 1–10%) and a small head, WBSNN scales with information: once the orbit dictionary has enough anchors, it matches or beats strong linear, kernel, tree, neural, and operator-learning baselines (e.g., DeepONet, FNO, NAIS-Net, EDMD). Grounded in operator theory and explicitly orbit-driven, WBSNN shines among operator-learning counterparts with consistent improvements. The richer $\alpha_{k,m}$ head is the geometry workhorse on nonlinear class boundaries, while the shared α_k head is the safer bias for noisy 1-D regressions and flip-noise regimes. Error-bar studies (20 seeds) show moderate spreads that contract as n or PCA retained variance grows; exact fits are preferred at small data or small d , while relaxed fits win as samples or horizons increase. In short, WBSNN learns the data’s geometry rather than memorizing examples—delivering competitive accuracy under CPU-only constraints.

4 Future Work.

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The present framework highlights two tightly linked open directions. Algorithmically, optimal scaffold selection remains a key challenge: given an operator family and dataset, how to efficiently construct orbit-constrained subspaces that minimize the projection error $\varepsilon_{\text{proj}}$ under computational and memory constraints, thereby collapsing the representational bottleneck of Phase 2 and enhancing the data efficiency of Weighted Backward Shift Realizer (WBSR) architectures.

Theoretically, the behavior of $\varepsilon_{\text{proj}}$ itself suggests a novel geometric structure detector, providing invariants of dataset–operator compatibility relative to orbit dynamics. Together, these directions aim toward a unified geometric theory of learnability in which operator realizability is governed by the intrinsic orbit geometry of the data, rather than by parametrization alone.

Extension to Infinite Dimensions. We established WBSNN’s foundations in infinite-dimensional sequence Banach spaces. Future work will systematically explore this regime on inherently infinite-dimensional datasets, leveraging WBSNN’s orbit dictionary to model complex, high-dimensional data structures while maintaining computational efficiency and interpretability.

Toward a Geometric Theory of Datasets. While this work introduces WBSNN and demonstrates its empirical and theoretical potential, we are still far from understanding how to optimally configure it for a given dataset $\mathcal{D}$. Interestingly, WBSNN consistently selects a vanishing regime ($\prod_{i=1}^d w_i < 1$) during Phase 1 optimization—promoting orbit stability and interpolation quality. This behavior, though not enforced, suggests the existence of dataset-specific configurations $(K_{\mathcal{D}}, W_{\mathcal{D}}, J_{\mathcal{D}})$ such that WBSNN acts as a functor extracting the intrinsic geometry or topology of $\mathcal{D}$. Such a perspective may ultimately allow us to reason about learning not only through data and models, but through the geometry they induce. If formalized, this could ground a geometric or category-theoretic framework for learning, where orbit representations encode generalization structure.

Acknowledgments I am especially grateful to Nicholas John Bordelon Jr. for believing in WB-SNNs even more than I did, and to Ozgur Martin for encouraging me to transition into the rich world of machine learning.

Reproducibility Statement An anonymous repository with all code, configs, and per-dataset notebooks/markdowns is available at https://github.com/wbsnn2025/wbsnn_for_reviewer.

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Appendix A. Proofs

Proof of Lemma 20. By Lemma 3, for every $m \geq 1$, there exists a scalar $\lambda_m > 0$ such that

$$W^{(m)}\varepsilon = \lambda_m \cdot W^{(m \bmod d)}\varepsilon.$$

Let $m = qd + r$, with $0 \leq r < d$ and $q \in \mathbb{N}$. Then, $\lambda_m = \left(\prod_{i=0}^{d-1} w_i\right)^q = \rho^q$. Hence,

$$\|W^{(m)}\varepsilon\| = \lambda_m \cdot \|W^{(r)}\varepsilon\| \leq \rho^q \cdot \max_{0 \leq r < d} \|W^{(r)}\varepsilon\| = \rho^{\lfloor m/d \rfloor} \cdot C,$$

where $C := \max_{0 \leq r < d} \|W^{(r)}\varepsilon\|$ depends only on ε and d . Since $\rho < 1$, it follows that $\|W^{(m)}\varepsilon\| \rightarrow 0$ exponentially as $m \rightarrow \infty$. ■

Proof of (14). Fix a tolerance $\varepsilon > 0$, let $\rho = \prod_{i=0}^{d-1} w_i$ and the effective horizon

$$M_{\text{eff}}(x; \varepsilon) := \min\{m \geq 0 : \|W^{(m)}J_k x\| \leq \varepsilon\}.$$

By Lemma 20, a sufficient condition for $\|W^{(m)}J_k x\| \leq \varepsilon$ is $C \rho^{\lfloor m/d \rfloor} \leq \varepsilon$, or equivalently $\rho^{\lfloor m/d \rfloor} \leq \frac{\varepsilon}{C}$. Since $0 < \rho < 1$, the function $x \mapsto \log_\rho x$ is decreasing; thus,

$\lfloor m/d \rfloor \geq \frac{\log(\varepsilon/C)}{\log \rho}$. Therefore,

$$m \geq d \left\lceil \frac{\log(\varepsilon/C)}{\log \rho} \right\rceil. \tag{17}$$

We have proved that if m satisfies (17), then $\|W^{(m)}J_k x\| \leq \varepsilon$. Thus, we arrive at the explicit upper bound $M_{\text{eff}}(x; \varepsilon) < d \left\lceil \frac{\log(\varepsilon/C)}{\log \rho} \right\rceil$. This completes the proof of (14). ■

Appendix B. Connection Between Theorem 4 and Algorithm ??

if this is not already mentioned in the intro like in realizability paper, then reduce this to half a page maximum

The pseudocode in Algorithm ?? is directly motivated by the constructive proof of Theorem 4, which guarantees that, for any dataset $\{(X_i, Y_i)\}_{i=1}^N \subset \mathbb{R}^d \times \mathbb{R}^d$ with $N \leq d$ and an appropriate choice of shift depths $L_i \in \{0, \dots, d-1\}$, there exists a linear map J such that $Y_i = JW^{(L_i)}X_i$ for all i . This result relies on the condition that the shifted vectors $\{W^{(L_i)}X_i\}$ are linearly independent.

In practice, checking exact linear independence (as required by condition (3)) would necessitate costly matrix rank computations or iterative orthogonalization methods such as Gram-Schmidt. Instead, our implementation employs a numerically efficient proxy: for each candidate point X_i , we select the shift depth L_i that minimizes the prediction error

$$L_i = \arg \min_{0 \leq L < d} \left| \tanh \left(\sum_j [W^{(L)}X_i]_j \right) - Y_i \right|.$$

Then, we accept $W^{(L_i)}X_i$ only if it lies outside the span of previously selected vectors, up to a norm threshold. This approximates the independence condition by ensuring that each new direction contributes novel information to the interpolation system.

Finally, the Phase 1 objective $\delta(W)$ measures the mean approximation error across all selected shifts and provides a practical surrogate for the deviation parameter δ defined in the isomorphism remark following Theorem 4. This aligns the empirical behavior of the model with the theoretical guarantees of exact interpolation.

In Phase 2, the interpolation operator J_k is computed via three modalities explained below.

Phase 2: Constructive Interpolation from Theorem 4 vs. Algorithm ?? The WBSNN framework supports three distinct strategies for constructing the interpolation operators J_k in Phase 2. Each reflects a different balance between theoretical guarantees, numerical stability, and practical considerations, and corresponds to a distinct interpretation of the core interpolation principle.

Constructive Interpolation via Rank-One Updates. Our constructive proof of Theorem 4 provides a direct algorithm for computing J_k using a sequence of *rank-one updates*. This approach builds J_k incrementally to ensure that each training pair $(X_i, Y_i) \in D_k$ is *exactly interpolated*, i.e.,

$$J_k W^{(L_i)}X_i = Y_i.$$

For each point, a linear functional f_i^* is constructed to vanish on previously selected directions and normalize $W^{(L_i)}X_i$. The update

$$J_k \leftarrow J_k + (Y_i - J_k W^{(L_i)}X_i) \otimes f_i^*$$

guarantees interpolation regardless of whether the matrix A_k is full-rank. This method is deeply aligned with the recursive structure of the interpolation theorem and is especially suitable when the independence condition can be enforced.

Closed-Form Interpolation via Pseudoinverse. In some exact interpolation experiments (e.g., IMDB), we instead construct J_k by solving the normal equations

$$J_k = (A_k^\top A_k)^{-1} A_k^\top B_k,$$

where $A_k \in \mathbb{R}^{|D_k| \times d}$ is the matrix of orbit vectors $W^{(L_i)}X_i$ and B_k contains their targets. This approach is efficient and widely used in regression problems, but *requires A_k to be full-rank* in order to guarantee exact interpolation. It serves as a computational baseline when rank conditions are known or likely to hold.

Regularized Least Squares for Non-Exact Interpolation. When the independence condition is relaxed (e.g., Gas Sensor Array Drift), we allow D_k subsets to include more or noisier points, possibly leading to rank-deficiency. To mitigate this, we solve a *regularized* least squares problem:

$$J_k = (A_k^\top A_k + \varepsilon I)^{-1} A_k^\top B_k.$$

This formulation does not guarantee exact interpolation but improves robustness and generalization, especially in noisy, compressed, or overparameterized regimes where exact inversion is unstable or ill-posed. It reflects a trade-off: sacrificing interpolation accuracy in favor of a better-conditioned solution and broader representational capacity.

These three approaches to Phase 2 span a spectrum of interpolation strategies. Rank-one updates are theoretically grounded and guarantee exact interpolation without requiring any rank condition, directly implementing the constructive proof of Theorem 4. The standard pseudoinverse provides the classical least-squares solution but assumes full column rank of the input matrix, performing best when the data is clean and well-conditioned. In contrast, the regularized pseudoinverse is robust to noise and rank deficiency, making it particularly effective in compressed or overparameterized settings where input representations may be low-dimensional or ill-conditioned. By integrating all three in our experiments, we not only validate the flexibility of WBSNN but also uncover structural differences in how interpolation affects generalization across datasets.

Implementation note. The pseudocode in Algorithm ?? adopts the regularized pseudoinverse formulation for Phase 2, as the constructive approach to exact interpolation is already formalized in Theorem 4. This complements the theoretical exposition while promoting numerical stability across diverse datasets. Ultimately, for optimal performance, the interpolation modality—rank-one update, pseudoinverse, or regularized pseudoinverse—should be selected based on dataset characteristics such as noise level, dimensionality, and rank structure, as each offers distinct trade-offs in generalization and robustness.

Table 1: Interpretability Details for WBSNN on Swiss Roll ($d = 3$, 120 samples, noise=0.5).

Phase 1 Points	1	Subset	Point X_1 : $[1.6078, 1.2816, 0.6559]$, Label: 1.0				
			Point X_2 : $[-2.0611, -1.0536, -0.3633]$, Label: 1.0				
			Point X_3 : $[0.4151, 1.5876, 0.5976]$, Label: 0.0				
			Point X_4 : $[-2.0067, -0.6176, -0.0628]$, Label: 1.0				
			Point X_5 : $[0.5160, -1.6329, -0.1805]$, Label: 0.0				
Phase 1 Weights			$W = [0.8, 0.8, 0.8]$, $\Delta = 3.1146$, $X_1 \in D_0, X_2 \in D_0, X_3 \in D_0, X_4 \in D_1, X_5 \in D_1$				
Phase 2 Interpolation			Point X_1 : $L_1 = 1$, Norm of $Y_1 - J_0 W^{(2)} X_1 = 7.77 \times 10^{-16}$				
			Point X_2 : $L_2 = 2$, Norm of $Y_2 - J_0 W^{(2)} X_2 = 6.66 \times 10^{-16}$				
			Point X_3 : $L_3 = 2$, Norm of $Y_3 - J_0 W^{(2)} X_3 = 7.39 \times 10^{-16}$				
			Point X_4 : $L_4 = 2$, Norm of $Y_4 - J_1 W^{(1)} X_4 = 8.88 \times 10^{-16}$				
			Point X_5 : $L_5 = 2$, Norm of $Y_5 - J_1 W^{(2)} X_5 = 7.24 \times 10^{-16}$				
Phase 3 + baselines			Model	Train Acc.	Test Acc.	Train Loss	Test Loss
			WBSNN	0.9868	0.9583	0.1011	0.1930
			Logistic Reg.	0.5921	0.7917	0.6636	0.6026
			Random Forest	1.0000	1.0000	0.5284	0.5685
			SVM (RBF)	0.9868	1.0000	0.3147	0.3330
			MLP 1HL	1.0000	1.0000	0.5134	0.5233
Held-out Point	Train	$X = [0.3204, -0.7333, 1.8079]$, Label: 1.0					
$\alpha_{k,m}(X)$ Matrix		$\begin{bmatrix} \alpha_{0,0}(X) & \alpha_{0,1}(X) & \alpha_{0,2}(X) \\ \alpha_{1,0}(X) & \alpha_{1,1}(X) & \alpha_{1,2}(X) \end{bmatrix} = \begin{bmatrix} 1.0000 & -1.0000 & 1.0000 \\ 0.5943 & -0.7987 & 0.2296 \end{bmatrix}$					
Prediction down	Break-down	Dim 1: $[J_0 W^{(0)} X, J_1 W^{(0)} X] = [0.0250, 0.0763]$, so we have Term $= [\alpha_{0,0}(X), \alpha_{1,0}(X)] \cdot [J_0 W^{(0)} X, J_1 W^{(0)} X]^T = 0.070374$.					
		Dim 2: $[J_0 W^{(1)} X, J_1 W^{(1)} X] = [-0.6677, -0.9095]$, so we have Term $= [\alpha_{0,1}(X), \alpha_{1,1}(X)] \cdot [J_0 W^{(1)} X, J_1 W^{(1)} X]^T = 1.394070$.					
		Dim 3: $[J_0 W^{(2)} X, J_1 W^{(2)} X] = [0.8810, -0.0718]$, so we have Term $= [\alpha_{0,2}(X), \alpha_{1,2}(X)] \cdot [J_0 W^{(2)} X, J_1 W^{(2)} X]^T = 0.864439$.					
		Sum of Terms: 2.328883, sigmoid probability $\sigma(2.328883) = 0.911241$, indicating high confidence in Class 1 due to positive terms, Pred Label: 1.0, True Label: 1.0.					
Test Point (Unseen)		$X = [0.4585, 0.6298, 0.0940]$, Label: 0.0					
$\alpha_{k,m}(X)$ Matrix		$\begin{bmatrix} \alpha_{0,0}(X) & \alpha_{0,1}(X) & \alpha_{0,2}(X) \\ \alpha_{1,0}(X) & \alpha_{1,1}(X) & \alpha_{1,2}(X) \end{bmatrix} = \begin{bmatrix} 0.0399 & -1.0000 & 0.1973 \\ 1.0000 & 0.8567 & 1.0000 \end{bmatrix}$					
Prediction down	Break-down	Dim 1: $[J_0 W^{(0)} X, J_1 W^{(0)} X] = [0.2504, -0.8192]$, so we have Term $= [\alpha_{0,0}(X), \alpha_{1,0}(X)] \cdot [J_0 W^{(0)} X, J_1 W^{(0)} X]^T = -0.809212$.					
		Dim 2: $[J_0 W^{(1)} X, J_1 W^{(1)} X] = [0.5423, -0.3129]$, so we have Term $= [\alpha_{0,1}(X), \alpha_{1,1}(X)] \cdot [J_0 W^{(1)} X, J_1 W^{(1)} X]^T = -0.810256$.					
		Dim 3: $[J_0 W^{(2)} X, J_1 W^{(2)} X] = [-0.2839, -0.8004]$, so we have Term $= [\alpha_{0,2}(X), \alpha_{1,2}(X)] \cdot [J_0 W^{(2)} X, J_1 W^{(2)} X]^T = -0.856379$.					
		Sum of Terms: -2.475847, sigmoid probability $\sigma(-2.475847) = 0.077569$, reflecting strong confidence in Class 0 via negative terms, Pred Label: 0.0, True Label: 0.0.					